

COURSE OUTLINE

Prepare surgical instruments and packs for surgery

Infection prevention

- Personal OT protective gears
- Sterilization
- Decontamination
- Waste management
- Aseptic techniques

- ☐ Types of surgical instruments
- ☐ Types of surgical packs and sets
- ☐ Types of surgery
- ☐ Sterilization in Operation Theatre
- ☐ Safety practices in Operation Theatre (assignment)
- ☐ Packing of surgical instruments
- ☐ Theatre layout
- ☐ Laminar flow in Operation Theatre

Participate in preparation of patient for surgery

- Basic Human anatomy and physiology
- ☐ ☐ Basic nursing skills
 - ☐ ☐ Identification and preparation of

Patient for surgery

- Patient positions
- ☐ ☐ Prepping and draping the patient
 - ☐ ☐ Basic communicable and noncommunicable

Diseases (assignment)

- ☐ ☐ Operation Theatre Ethics
- ☐ ☐ Maintaining Operation Theatre

Basic Human anatomy and physiology

☐ ☐ Basic nursing skills

DISPOSAL OF THEATRE WASTE

- Types of wastes in Operation Theatre

☐ ☐ Waste disposal methods

☐ ☐ Handling of Operation Theatre waste

INTRODUCTION TO PREVENTION OF NOSOCOMIAL INFECTION 1

NOSOCOMIAL INFECTIONS

- They are also called hospital – acquired infections
- They are infections acquired in the hospital or other health care facility that were not present or incubating during the time the patient was admitted
- Include those infections that become symptomatic after the patient has been discharged as well as among medical personnel
- Most of these infections are transmitted by medical personnel who fail to practice proper hand washing procedures or change of gloves between client contact

Nosocomial pathogens

They include; bacteria, virus, fungi, parasites

Common nosocomial infections include the following;

1. Urinary tract infections (UTIs): UTIs are one of the most common nosocomial infections.

They occur when bacteria, typically *Escherichia coli*, enter the urinary tract through catheters or other medical devices. Symptoms may include frequent urination, burning sensation, and cloudy urine.

2. Surgical site infections (SSIs): SSIs occur after surgery and can affect the incision site or deeper tissues. They are caused by bacteria entering the surgical site during the procedure or postoperatively. Risk factors include poor surgical techniques, prolonged surgery, and compromised immune system.

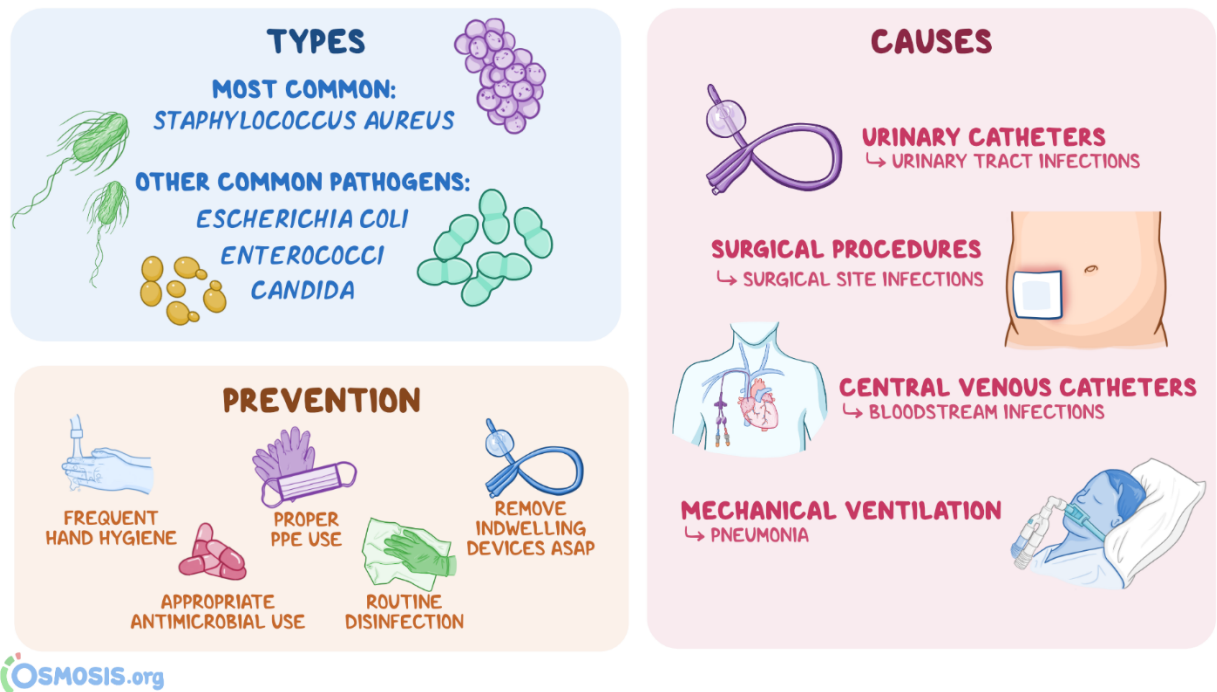
3. Pneumonia: Hospital-acquired pneumonia (HAP) is a lung infection that develops after being hospitalized for at least 48 hours. It can be caused by bacteria, viruses, or fungi. Ventilator-associated pneumonia (VAP) is a type of HAP that occurs in patients on mechanical ventilation.

4. Bloodstream infections: These infections, also known as sepsis or bloodstream infections (BSIs), occur when pathogens enter the bloodstream. Central line-associated bloodstream infections (CLABSI) are a common type, caused by bacteria colonizing central venous catheters.

5. Clostridium difficile infection (CDI): CDI is a gastrointestinal infection caused by the bacterium Clostridium difficile. Patients on antibiotics are at higher risk, as the medications can disrupt the normal gut flora, allowing C. difficile to flourish. Symptoms range from mild diarrhea to severe colitis.

6. Methicillin-resistant Staphylococcus aureus (MRSA) infections: MRSA is a type of bacteria that is resistant to many antibiotics. It can cause various infections, including skin and soft tissue infections, surgical site infections, and bloodstream infections. MRSA is a significant concern in healthcare settings due to its resistance and ability to spread easily.

1. Central line-associated bloodstream infections (CLABSI)- blood transfusion tubes, injectable and contaminated blood.
2. Catheter-associated urinary tract infections (CAUTI)- urinary catheters
3. Surgical site infections (SSI) - improper care of surgical wounds.
4. Ventilator-associated pneumonia (VAP) – non-sterile ventilators, contaminated air.



The 10 Standard Infection Control Precautions (SICP)

SICPs are the essential standards of infection prevention and control measures taken by healthcare facility staff (normally in advance) to reduce the risk of spreading infections.

1. Patient assessment for infection risk
2. Hand hygiene
3. Respiratory and cough hygiene
4. Personal protective equipment (PPE)
5. Safe management of equipment
6. Safe management of environment
7. Safe management of blood and body fluids
8. Safe management of linen

9. Safe disposal of waste (including sharps)

10. Occupational safety & Exposure

INFECTION PREVENTION

Personal OT protective gears

Gloves – protect hands and allow efficient removal of organisms from hands

Gowns and Aprons – protect skin and clothing

Face masks– protect mucous membranes of mouth and nose

Respirators- prevent inhalation of infectious material

Goggles – protect eyes

Face shields – mucous membranes of face, mouth, nose and eyes

Sterilization

Sterilization is a process which achieves the complete destruction or killing of all microorganisms, including bacterial spores.

Sterilization is principally accomplished by:

- Steam under pressure (Autoclave)
- Dry heat (Hot air oven)
- Chemical agents such as ethylene oxide or low temperature methods
- Chemical sterilant as glutaraldehyde 2 %

Preferred methods of sterilisation

Table 2: Various sterilisation methods with their advantages, disadvantages and preferred use				
Method	Time	Advantage	Disadvantage	Suitable for
Autoclave	~ 45 min	Kill spores, can sterilise liquids	Items must be heat and moisture resistant	Gowns, dressings, drapes, Metal instruments
Hot air oven	1.5 to 2 hrs	Not much pressure build up	Prions not killed every time	Metal instruments, swabs
Ethylene oxide	5 psi- 12 hrs 10 psi- 6 hrs	Low temperature	Explosive, long duration	Heat labile tubings, cryo and laser probes
Plasma system	Short cycle 74 min.	Low temperature	Specific packaging needs	Same as ETO
Gluteraldehyde	Vegetative pathogens- 15 min. Spores- 3 hrs	For heat labile equipments	Irritant	Metal instruments, Plastic Glass

➤ Steam Under Pressure

Autoclave: Autoclave sterilization is safe method of sterilization; it kills bacteria, spores, viruses and fungus. For this, all items are arranged inside autoclave in a way that allows steam to circulate freely. STACKING must not be done. Gloves must be worn while handling the instruments to avoid infective material & cuts. Then the time duration, temperature and pressure is adjusted. Timing should begin after the autoclave reaches the desired temperature and pressure. Autoclave should be opened once pressure gauge reads “o”. All items must be dried before use. Wet packs must be considered non-sterile. Labelling must be accurate with contents, date of processing and expiration date.

Autoclaved instruments should be used within 48 hours. The water should be drained out daily to avoid settling of salt on the instruments and in the chamber. The autoclave should be serviced once in 6 months

Flash or high speed autoclave: For Flash or high speed autoclave the instruments are first cleaned in clean sterile water or distilled water. These are type of gravity downward displacement autoclaves, which fills with steam and displaces air downward and forces it out of drainage valve. The canulas

should be cleaned separately with RL solution. The cleaning water should be changed after cleaning 4 or 5 sets.

B. Dry heat sterilisation

Less preferred in recent era, but indicated for reusable glass, metal instruments, oil, ointments and powders. All items should be cleaned and dried and then put in metal container and heated upto holding temperature (141-180°) and kept for recommended time (3 hrs to 30 min.). items are allowed to cool inside oven up to room temperature and then can be used or stored.

C. Low temperature sterilization

This is used for heat- and moisture sensitive medical devices. The types include:

- ETO sterilization
- Plasma sterilization

ETO (Ethylene oxide) sterilization: This is an alkylating agent which reacts with DNA and destroys the ability of microorganisms to metabolize or reproduce. This was a method of choice for many heat- and moisture-sensitive devices but new methods of low temperature sterilization are now often used as substitutes. Effective EO sterilization depends on concentration of sterilant, relative humidity, temperature and exposure time. For ETO sterilisation after loading the steriliser, air is removed with vacuum followed by heating upto 45-55°C with relative humidity of 60%. Required exposure to ETO is upto 12 hours if 5psi and 6 hrs if 10 psi subsequently gas removal by vacuum and flushing with air 4 times.

Hydrogen peroxide Plasma sterilisation: Plasma is defined as highly ionised

gas composed of ions, electrons, and neutral particles. Plasmas are generated by introducing a precursor gas or vapor (e.g., hydrogen peroxide or peracetic acid) into a chamber under low-vacuum conditions and then exciting the gas or vapour with microwave or radiofrequency energy. The cycle time is approximately 75 minutes. They have special packaging needs as Cellulose-containing wrappers are incompatible with hydrogen peroxide processes because they absorb the peroxide and do not allow effective penetration. Commercially available nonwoven polypropylene wraps and poly olefin pouches are material of choice.

D. Chemicalsterilisation

a) Keep sharp instruments in activated solution of Cidex (2% Glutaraldehyde) for 15 minutes

b) Wash them 3 times with sterile water, kept in the three trays.

c) Once mixed, they are usually sterile for up to 14 days.

d) Replace solutions any tim Y
e they become cloudy.

Other Specifications

1. All items must be labelled with the date of sterilisation.
2. Every sterilised item has an expiry date.
3. Items must be inspected before use to ensure that there are no tears, punctures, open seams, moisture, soiling from being dropped on the floor, etc.
4. Once a package has been opened, it is no longer considered sterile, whether or not its contents are used.
5. Don't store non-sterile items along with sterile items.
6. If the sterility of an item cannot be assessed, it must be re-sterilised or discarded.

DECONTAMINATION

1. **Preparation:**

- Ensure a designated decontamination area with separate zones for dirty, clean, and sterile instruments.
- Establish a workflow system for the handling of instruments, including designated areas for sorting, cleaning, inspection, and sterilization.
- Implement standard operating procedures (SOPs) for decontamination processes and ensure staff are trained on proper techniques.
- Verify that all equipment and cleaning solutions are functioning properly and within expiration dates.

2. **Disassembly:**

- Disassemble instruments according to instrument-specific guidelines provided by the manufacturer.
- Separate delicate instruments from heavy or sharp instruments to prevent damage during cleaning.
- Inspect instruments for any visible damage or wear that may require repair or replacement.

1. **Preparation:**

- Gather all theatre instruments that require disinfection.
- Wear appropriate personal protective equipment (PPE) such as gloves, gown, and eye protection.

2. **Cleaning:**

- Start by removing any visible debris or organic material from the instruments using a brush or enzymatic cleaner.
- Clean the instruments thoroughly with a detergent solution to remove dirt and grime.
- Rinse the instruments with clean water to remove detergent residue.

3. **Selection of Disinfectant:**

- Choose an appropriate disinfectant based on the type of instruments and the level of disinfection required.
- Ensure that the disinfectant is compatible with the materials of the instruments.

4. **Preparation of Disinfectant:**

- Follow the manufacturer's instructions to prepare the disinfectant solution at the correct dilution.
- Use room temperature water unless specified otherwise by the manufacturer.

5. **Immersion:**

- Immerse the cleaned instruments in the disinfectant solution, ensuring all surfaces are completely submerged.
- Ensure that the instruments are not overcrowded in the disinfectant solution to allow for effective contact.

6. **Contact Time:**

- Allow the instruments to remain in the disinfectant solution for the recommended contact time specified by the manufacturer.

- This contact time is crucial to ensure that the disinfectant can effectively kill microorganisms.

7. **Rinsing:**

- After the specified contact time, remove the instruments from the disinfectant solution.

- Rinse the instruments thoroughly with sterile water to remove any residual disinfectant.

8. **Drying:**

- Allow the disinfected instruments to air dry completely in a clean and designated area.

- Ensure that the instruments are dried thoroughly before storage to prevent microbial growth.

9. **Verification:**

- Inspect the instruments to ensure they are visibly clean and free from any residues.

- Check for any signs of damage or corrosion that may have occurred during the disinfection process.

10. **Storage:**

- Store the disinfected instruments in a clean, dry, and well-ventilated area.

- Avoid overcrowding and ensure proper organization to prevent contamination.

11. **Documentation:**

- Maintain detailed records of the disinfection process for each set of instruments, including the type of disinfectant used, contact time, and date of disinfection.
- Follow institutional protocols for documentation and record-keeping.

3. **Cleaning:**

a. **Manual Cleaning:**

- Pre-rinse instruments in cold water to remove gross debris and blood.
- Immerse instruments in an enzymatic detergent solution and scrub with a soft brush to remove organic material.
- Pay attention to joints, hinges, and serrated edges where debris can accumulate.

b. **Ultrasonic Cleaning:**

- Place disassembled instruments in an ultrasonic cleaner filled with an enzymatic solution.
- Operate the ultrasonic cleaner according to manufacturer's instructions for the recommended time and frequency.

4. **Rinsing:**

- Thoroughly rinse instruments with clean water to remove detergent residue and loosened debris.
- Use distilled or filtered water to prevent water spots and mineral deposits on instruments.

5. **Drying:**

- Dry instruments using a clean, lint-free cloth or air dryer to prevent water staining and corrosion.
- Consider using a drying cabinet or warm air sterilizer for efficient and controlled drying.

6. **Inspection:**

- Conduct a visual inspection of each instrument for cleanliness, damage, and functionality.
- Use magnification or illuminated inspection tools to ensure all surfaces are free of debris.
- Check for signs of corrosion, discoloration, or wear that may affect instrument performance.

7. **Packaging:**

- Place cleaned and inspected instruments in sterilization pouches, trays, or containers suitable for the sterilization method.
- Ensure proper assembly of instrument sets and trays according to surgical requirements.
- Label packages with appropriate information, including date of processing, contents, and sterilization indicators.

8. **Sterilization:**

- Select the appropriate sterilization method based on instrument material, size, and intended use.
- Follow sterilization parameters for autoclaving, chemical sterilization, or other methods as recommended by instrument manufacturers.
- Monitor and record sterilization cycles, including time, temperature, pressure, and sterilization indicators.

9. **Storage:**

- Store sterilized instruments in a clean, dry, and well-ventilated area with controlled humidity levels.
- Keep instrument sets sealed and protected from environmental contaminants until ready for use.
- Rotate instrument sets regularly to ensure proper air circulation and prevent damage from prolonged storage.

10. **Documentation:**

- Maintain detailed records of the decontamination process, including cleaning methods, sterilization cycles, and instrument maintenance.
- Document any deviations from standard procedures, equipment malfunctions, or quality control issues.
- Retain records for a specified period as required by regulatory guidelines and facility policies.

certainly! Here is a step-by-step process for donning (putting on) and doffing (taking off) personal protective equipment (PPE) properly:

Donning PPE:

1. **Hand Hygiene**:

- Wash hands with soap and water or use hand sanitizer before starting the donning process.

2. **Gown**:

- Put on a gown by sliding your arms into the sleeves and tying the gown securely at the back of your neck and waist.

3. **Mask**:

- Place the mask over your nose and mouth, securing it behind your ears or tying it securely at the back of your head.

4. **Goggles/Face Shield**:

- Put on goggles or a face shield, ensuring it covers your eyes and the sides of your face.

5. **Gloves**:

- Put on gloves, ensuring they cover the wrists of the gown.

Doffing PPE:

1. **Gloves**:

- Grasp the outside of one glove at the wrist and peel it off, turning it inside out.
- Hold the removed glove in the gloved hand.
- Slide fingers of the ungloved hand under the remaining glove at the wrist and peel it off, turning it inside out.
- Discard both gloves in a designated receptacle for contaminated items.

2. **Gown**:

- Unfasten the gown ties at the back of the neck and waist, avoiding contact with the outside of the gown.
- Peel the gown away from your body, rolling it inside out as you remove it.
- Discard the gown in a designated receptacle for contaminated items.

3. **Hand Hygiene**:

- Perform hand hygiene using soap and water or hand sanitizer.

4. **Goggles/Face Shield**:

- Remove goggles or face shield by handling the headband or earpieces.
- Avoid touching the front of the goggles or shield.
- Clean and disinfect reusable goggles or face shields as per facility guidelines.

5. **Mask**:

- Remove the mask by handling the ear loops or ties, avoiding touching the front of the mask.

- Discard disposable masks in a designated receptacle.
- If using a reusable mask, follow facility guidelines for cleaning and storing.

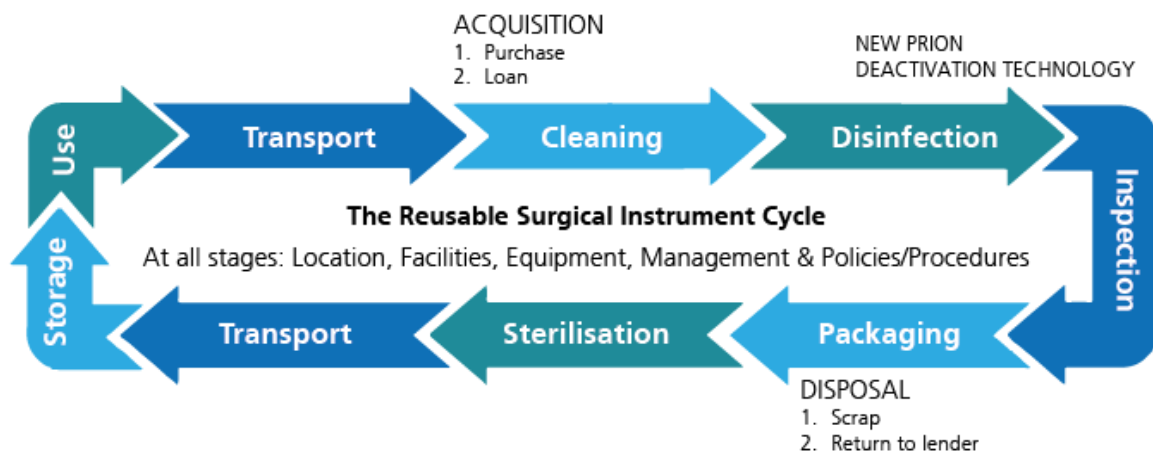
Additional Tips:

- Always follow facility-specific protocols for donning and doffing PPE.
- Perform a fit check to ensure PPE is worn correctly.
- Avoid touching your face, hair, or other surfaces while wearing PPE.
- Dispose of PPE in designated receptacles for contaminated items.
- Wash hands thoroughly after doffing PPE.

By following these step-by-step instructions for donning and doffing PPE, healthcare workers can help minimize the risk of contamination and protect themselves and others in healthcare settings.

This is the process of removing or neutralizing contaminants that have accumulated on personnel and equipment.

Decontamination involves 3 stages;



General cleaning,

- Ensure all civil work is completed
- Ensure all movable equipments have been shifted
- Wear utility gloves
- Wipe all surfaces of the OT using water and soap. Repeat until all visible dust is removed.
- Allow all surfaces to dry completely.
- Wipe all surfaces (including ceiling and walls up to the height the hands can reach)
- Allow to dry completely.

Disinfection

- Fog the OT with high level disinfectant until a fog is seen in the air.
- Stop and remove the fogger and close the OT for at least 1 hour with any ventilation system/AC off.
- After 1-2 hours open the OT and take post fogging swabs. change the OT dress, cap, mask and use sterile gloves when performing the sampling.
- In this case, only the person taking the samples should enter the OT.

Sterilization

- Involves dry heat sterilization
- Steam under pressure sterilization
- Low temperature sterilization

WASTEMANAGEMENT (31st jan)

Waste- any substance that is discarded after primary use or is worthless, defective and of no use.

Waste management –collection, transportation, handling and disposal process of dealing with removal of human waste.

Types of waste

Waste and by-products cover a diverse range of materials, as the following list illustrates:

- **Infectious waste:** waste contaminated with blood and other bodily fluids (e.g. from discarded diagnostic samples), cultures and stocks of infectious agents from laboratory work (e.g. waste from autopsies and infected animals from laboratories), or waste from patients with infections (e.g. swabs, bandages and disposable medical devices);
- **Pathological waste:** human tissues, organs or fluids, body parts and contaminated animal carcasses;
- **Sharps waste:** syringes, needles, disposable scalpels and blades, etc.;
- **Chemical waste:** for example solvents and reagents used for laboratory preparations, disinfectants, sterilants and heavy metals contained in medical devices (e.g. mercury in broken thermometers) and batteries;
- **Pharmaceutical waste:** expired, unused and contaminated drugs and vaccines;
- **Cytotoxic waste:** waste containing substances with genotoxic properties (i.e. highly hazardous substances that are, mutagenic, teratogenic or carcinogenic), such as cytotoxic drugs used in cancer treatment and their metabolites;
- **Radioactive waste:** such as products contaminated by radionuclides including radioactive diagnostic material or radiotherapeutic materials; and
- **Non-hazardous or general waste:** waste that does not pose any particular biological, chemical, radioactive or physical hazard.

The major sources of health-care waste are:

- hospitals and other health facilities
- laboratories and research centres
- mortuary and autopsy centres

- animal research and testing laboratories
- blood banks and collection services
- nursing homes for the elderly

Stages involved in waste management

1. Handling

- All staff members handling waste products should receive periodic reminders and refresher training that includes;
- Information on the techniques and risks associated with the handling of waste,
- Procedures for dealing with spillages.
- Other accidents and instructions on the use of protective clothing.

2. **segregation** – involves separation of healthcare waste according to their categories and labeling waste containers. It is the most important step to reduce the risk and amount of hazardous waste

3. **Collection** – proper ppe's and waste transportation utility supply such as waste bins , trolley and wheelbarrow are vital in waste collection.

4. Storage

- The waste storage areas must be completely enclosed and separate from supply rooms and food preparation areas.
- Waste must always be segregated into different fractions based on their potential hazard and disposal route.
- The segregation of waste items is the responsibility of the handler. When stored, waste should be monitored closely and removed in a timely manner.
- All waste containers should be clearly labeled and hazardous and non-hazardous wastes should never be mixed.

5. **Transportation**- Proper personal protective equipments supplies and easily cleanable sealed containers such as plastic buckets or trolleys can be used to transport waste from their temporary storage site to their treatment/permanent disposal.

6.Treatment

There are five basic processes for the treatment of hazardous components in health-care waste, specifically, sharps, infectious and pathological wastes: thermal, chemical, irradiation, biological and mechanical.

These processes include; chemical, thermal, biological, and physical methods.

- 7.**Disposal** – Dispose waste in accordance with the requirements of applicable national/ international regulations. Establish a system of separate disposal of the different types of waste.

ASEPTIC TECHNIQUE

Aseptic technique is a standard healthcare practice that helps prevent the transfer of pathogens to or from an open wound and other susceptible areas on a patient's body.

Aseptic technique helps prevent healthcare-associated infections (HCAIs). An HCAI is an infection that a person acquires as a result of treatment from a healthcare professional.

Common HCAIs include:

- catheter-associated urinary tract infection
- central line-associated bloodstream infection – catheter used put on veins of arm or neck region to administer drugs.
- *Clostridium difficile* infection - an **infection** of the large intestine (colon).
- surgical site infection
- ventilator-associated pneumonia -Ventilator-associated pneumonia is a lung infection that develops in a person who is on a ventilator. A ventilator is a machine that is used to help

a patient breathe by giving oxygen through a tube placed in a patient's mouth or nose, or through a hole in the front of the neck

aseptic technique is used by health care providers when;

- .performing surgical procedures
- performing biopsies
- dressing surgical wounds or burns
- suturing wounds
- inserting a urinary catheter, wound drain, intravenous line, or chest tube
- administering injections
- using instruments to conduct a vaginal examination
- delivering babies

Principles of aseptic technique

Table 1. Principles of aseptic technique

Action	Rationale
Hand hygiene	Remove transient micro-organisms from the hands
Safe storage of equipment	Prevent damage to the sterile equipment, preserve sterility of the equipment and prevent microbial contamination
Cleaning of the procedure trolley or tray	Reduce microbial contamination
Preparation of equipment	Prevent microbial contamination of sterile equipment
Personal protective equipment (PPE)	Aprons provide protection from potential contamination from the healthcare worker's (HCW) uniform and the procedure and also protects the HCW from potential contamination from the procedure Non-sterile gloves provide protection for the HCW from contamination from blood and body fluids that may contaminated the hands Sterile gloves protect key sites from potential microbial contamination from the HCW's hands
Preparation of the environment	Reduce microbial contamination during the procedure
Preparation of the patient	Gain informed consent and reduce anxiety
Waste disposal	Prevent contamination of the environment
Documentation	Provide essential communication and meet the standards in the NMC Code (2015)

Preparing the Skin

- Clean the site with soap and water.
- Wipe the skin in a circular motion, beginning in the center of the site then moving outwards.
- ☐ Use sterile cotton balls or cotton wool sponges held on sponge forceps.
- Any of the following antiseptics may be used as listed in the preferred order:
 - An iodophor (Betadine); then wait for two minutes and wipe off the excess with sterile dry cotton or gauze

- 4% Chlorhexidine (Hibiclens) wipe off excess with sterile, dry cotton or gauze
- 1-3 % iodine, followed by 60-90% alcohol (ethyl or isopropyl) then allow to air dry
- Chlorhexidine with cetrimide (Savlon) wipe off the excess with sterile, dry cotton or gauze

TYPES OF SURGICAL INSTRUMENTS

Surgical instruments may be categorized according to their functions as follows;

1. Cutting and dissecting instruments: used for incision/cutting of tissues

They include; Scalpels, scissors, saws, Elevators, and diathermy/cautery.

2. Grasping or holding instruments: used to hold incisions open or hold an organ (or tissue) during surgical procedures.

- They included **forceps** and **clamps**.

3. Hemostatic instruments: This includes instruments utilized for the cessation of bleeding

- **Artery forceps, Sutures and Cautery.**

4. Retractors:- **surgical instrument used to separate the edges of a surgical incision or wound**, or to hold back underlying organs and tissues so that body parts under the incision may be accessed. These can broadly be hand held (often by a junior assistant) or self retaining. **Hand held include;**

- Army- Navy - Used to retract shallow or superficial incisions. ...
- Ribbon (Malleable) Used to retract deep wounds. ...
- Hohmann - used in orthopedics to expose bone for procedures.
- Farabeuf - general use.

Self retaining retractors include;

Weitlaner and Gelpie retractors- These are used to retract skin edges for superficial procedures.

5. Tissue unifying instruments and materials:

- This would include instruments that aid in tissue unification. They include; needle holders and staple applicators)

Surgical instruments packs and sets.

Classification of sets

Specific set

General set

Guidelines when handling sets

- Inspect each instrument before and after use to detect imperfections
- Set aside damaged instruments and send them for repair or replacement promptly
- Use instruments for the purpose for which they are designed
- Handle instruments gently
- Provide protection for delicate instruments
- Clean instruments meticulously
- Keep oil away from instruments
- Give instruments regular maintenance
- Take inventory
- Rotate sterile instruments in order to use them according to expiring date.
- A.V. Fistula Surgical Instrument Set. - Surgical connection of an artery to a vein.
- Abdominal Hysterectomy Instrument Set. - **a surgical procedure that removes the uterus through an incision in lower abdomen**

- Abdominoplasty Surgical Instrument Set. - **a cosmetic surgical procedure to improve the shape and appearance of the abdomen**
- Amputation Surgical Instrument Set. - **surgical removal of part of the body**, such as an arm or leg.
- Anterior & Posterior (A&P) Repair Surgical Instrument Set. - **anterior repair refers to correction of the front wall of the vagina**; and posterior repair refers to correction of the back wall of the vagina
- Bladder Surgical Instrument Set.- surgery made in the bladder.
- Blepharoplasty Surgical Instrument Set. – used in **surgery that repairs droopy eyelids**
- **Breast augmentation surgical instruments set** - surgery to increase breast size.
- **Breast reduction surgical instruments**- surgical procedures to reduce size of breasts
- **Cardiovascular surgical instruments pack** - any surgical procedure that involves the heart, or the blood vessels that carry blood to and from the heart.
- **Foot and ankle surgical instruments set** – surgical procedures of foot and ankle
- **Cesarean section surgical instruments set**- A cesarean birth is a surgical procedure used to deliver a baby when a vaginal delivery cannot be safely done
- **Gastrointestinal surgical instruments set** - ***Gastrointestinal surgery*** is a treatment for diseases of the parts of the body involved in digestion.
- **Hip replacement instruments set** – surgical procedures involving hip replacement
- **Knee surgery instruments set** – surgery of the knees
- **Infant laparotomy instrument set** - surgical incision into the abdominal cavity performed to examine the abdominal organs and aid diagnosis of any problems.
- **Open heart surgical instruments set** - any type of ***surgery*** where the chest is cut ***open*** and ***surgery*** is performed on the muscles, valves, or arteries of the ***heart***.
- **Ophthalmology surgical instruments** - diagnosis and treatment of eye disorders

Types of surgery

Surgery may be categorized according

Urgency

Elective- Does not require urgent interventions, it is planned and performed when patient chooses

Urgent –Necessary for patient's health.it is indicated to prevent additional health problems.

Emergency-performed immediately to save the individual lives or preserve function of a body part

Required-patient needs to have surgery, plan within a few weeks or months ie prostatic hyperplasia without bladder obstruction

Optional-decision rests with patient and it is personal preference

Based on body part

- Appendectomy. An appendectomy is the surgical removal of the appendix, a small tube that branches off the large intestine, to treat acute appendicitis. Appendicitis is the acute inflammation of this tube due to infection.
- Breast biopsy. A biopsy is a diagnostic test involving the removal of tissue or cells for examination under a microscope. This procedure is also used to remove abnormal breast tissue.
- **Carotid endarterectomy**. Carotid endarterectomy is a surgical procedure to remove blockage from carotid arteries, the arteries located in the neck that supply blood to the brain.
- **Cataract surgery**. Cataract surgery involves the removal of the cloudy lens, which is replaced with a clear artificial lens implant.
- Cesarean section (also called a c-section). Cesarean section is the surgical delivery of a baby by an incision through the mother's abdomen and uterus. This procedure is done when doctors determine it a safer alternative than a vaginal delivery for the mother, baby, or both.
- Cholecystectomy. A cholecystectomy is surgery to remove the gallbladder (a pear-shaped sac near the right lobe of the liver that holds bile). A gallbladder may need to be removed if the organ is prone to troublesome gallstones, if it is infected, or becomes cancerous.
- Coronary artery bypass. This surgery is often done in people who have chest pain and coronary artery diseases. During the surgery, a bypass is created by grafting a piece of a

vein above and below the blocked area of a coronary artery, enabling blood to flow around the obstruction.

- **Debridement of wound, burn, or infection.** Involves the surgical removal of foreign material and/or dead, damaged, or infected tissue from a wound or burn. By removing the diseased or dead tissue, healthy tissue is exposed to allow for more effective healing.
- **Dilation and curettage (also called D & C).** A D&C is a minor operation in which the cervix is dilated (expanded) so that the cervical canal and uterine lining can be scraped with a curette (spoon-shaped instrument).
- **Free skin graft.** A skin graft involves detaching healthy skin from one part of the body to repair areas of lost or damaged skin in another part of the body. Skin grafts are often done as a result of burns, injury, or surgical removal of diseased skin. They are most often done when the area is too large to be repaired by stitching or natural healing.
- **Hemorrhoidectomy.** A hemorrhoidectomy is the surgical removal of hemorrhoids, distended veins in the lower rectum or anus.
- **Hysterectomy.** A hysterectomy is the surgical removal of a woman's uterus. This may be done through an abdominal incision or vaginally. The ovaries may be removed at the same time.
- **Hysteroscopy.** Hysteroscopy is a surgical procedure used to help diagnose and treat many uterine disorders. The hysteroscope (a viewing instrument inserted through the vagina for a visual exam of the canal of the cervix and the interior of the uterus) can transmit an image of the uterine canal and cavity to a television screen.
- **Inguinal hernia repair.** Inguinal hernias are when the small intestine bulges through a weak area in the lower abdominal muscles. An inguinal hernia occurs in the groin. Surgical repair pulls the intestine back to its original location.
- **Low back pain surgery.** The type of surgery done on the back due to low back pain and it depends with a type of diagnosis.
- **Mastectomy.** A mastectomy is the removal of all or part of the breast. Mastectomies are usually done to treat breast cancer.
- **Partial colectomy.** A partial colectomy is the removal of part of the large intestine (colon) which may be done to treat cancer of the colon or inflammatory conditions.

- **Prostatectomy.** The surgical removal of all or part of the prostate gland and urethra. A prostatectomy may be done for an enlarged prostate, or if the prostate gland is cancerous.
- **Tonsillectomy.** The surgical removal of one or both tonsils. Tonsils are located at the back of the mouth and help fight infections.

An operating theatre, operating room, surgery suite or a surgery centre is a room within a hospital within which surgical and other operations are carried out.

OT are designed and built to carry out investigative, diagnostic, therapeutic and palliative procedures of varying degrees of invasiveness. Many such setups are customized to the requirements based on size of hospital, patient turnover and may be speciality specific.

The aim is to provide the maximum benefit for maximum number of patients arriving to the operation theatre. Both the present as well as future needs should be kept in mind while planning OT.

OT design

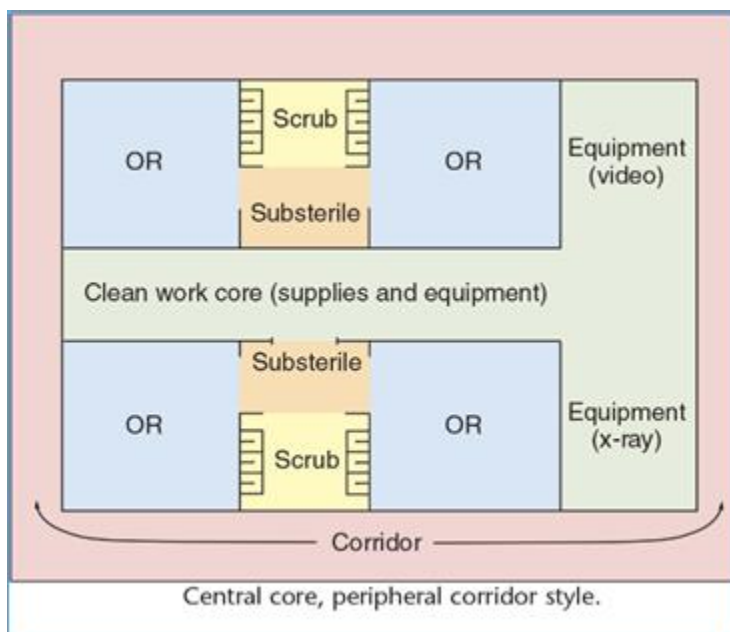
Several basic design types are used in surgical services departments, depending on the age of the facility and the physical design of the areas outside the department. All surgery departments are designed with the idea of controlling traffic patterns and quickly providing each operating room (OR) with the necessary supplies during and after each case, while keeping clean and contaminated traffic patterns separate.

Most surgical suites are constructed according to a variation of one or more basic designs. The basic designs are:

- **Central corridor plan** – these plans are two types
 - ✓ **Racetrack plan** - In this design, the front entrance to each OR is from the outer corridor, and supplies are retrieved through a rear entrance to the room leading to the central-core storage and work areas.

- ✓ **Hotel plan** - In this design, the ORs are situated along a central corridor, with separate clean core and soiled work areas. The primary difference in this plan is that all traffic enters and exits the surgery department through a single entrance or a primary entrance and holding area entrance situated along the same corridor.

- **Peripheral corridor** - In this design, the front entrance to each OR is from the peripheral corridor, and supplies are retrieved through a rear entrance from the OR leading to the central-core storage and work areas.



- **Specialty grouping plan** - The "specialty grouping" plan is simply a variation on the hotel or race track plan, in which ORs are grouped by specialty (e.g., neurosurgery, general surgery), each with its own closely associated clean storage areas and, in some cases, each with its own soiled instrument work area.

Different Zones/ Area.

The OR department is divided into four zones based on varying degrees of cleanliness, in which the bacteriological count progressively diminishes from the outer to the inner zones (operating

area) and is maintained by a differential decreasing positive pressure ventilation gradient from the inner zone to the outer zone.

1. Unrestricted zone or clean area.
2. Semi restricted zone or sub sterile area
3. Restricted zone or sterile area
4. Disposal zone or area

1. **Unrestricted/ Clean zone or clean area (next lesson 15th)**

- This is the area where hospital personnel, OT personnel, patient's & their attendants can move about in street clothes.
- Connects protective zone to aseptic zone and has other areas also like
 - Stores & cleaner room
 - Equipment store room
 - Maintenance workshop
 - Kitchenette (pantry)
 - Firefighting device room
 - Emergency exits
 - Service room for staff
 - Close circuit TV control area

2. **Semi restricted/ Protective zone or sub-sterile area**

- After changing from street clothes, with clean gown, cap & OT slipper, the OT team or personnel are, enter in this area. (Operating room attire is required)
- It includes -
 - Change rooms for all medical and paramedical staff with conveniences
 - Transfer bay for patient, material & equipment's
 - Rooms for administrative staff
 - Stores & records
 - Pre & post-operative rooms
 - I.C.U. and P.A.C.U.

- Sterile stores

3. **Restricted/Aseptic zone or sterile area:**

- This zone has operation theatres & operation room where operations are done.
- This is a place where staff & patients in street clothes & shoes are not allowed to enter. Scrubbed personnel wear sterile gowns & gloves & get ready for operation procedure. (Operating room attire is required)

4. **Disposal zone or area**

This is where all theatre waste is temporarily disposed before its being transported to permanent disposal points.

- Dirty utility area
- Disposal corridor
- OT attire mandatory in this area

Advantages of zoning

- Minimizes risk of hospital infection.
- Minimizes unproductive movement of staff, supplies & patient.
- Increases efficacy of operative team members.
- Ensures smooth workflow.
- Decreases hazards in operating room.
- Ensures proper positioning of equipment's.

Sub areas

- **Nurses station or counter** - Nurses station is an area where nurses and other health care staff sit behind when not working directly with patients and can perform some of their duties. The station has a counter that can be approached by visitors and patients who wish to receive attention from the nurses. It should be spacious enough to allow two or more personnel to work together.
- **Pre-operative check area (reception)** - It is the area where OT nursing personnel receive patients. Here patients and his records are checked.

- **Holding area** - This area is planned for IV line insertion, preparation, catheter / gastric tube insertion, connection of monitors, & shall have O2 and suction lines. Facility for CPR should be available in this area.
- **Induction/Anaesthetic room** It provides space for anaesthetic trolleys and equipment and is located with direct access to circulation corridors and ready access to the operating room. It also allow cleaning, testing and storing of anaesthesia equipment. It should have sufficient power outlets and medical gas panels for testing of equipment.
- **Post anaesthetic care units (PACU)/Post-operative or recovery room** – Room where patients are kept & cared until they are out of anaesthesia & until their vital signs become stable. These should contain a medication station, hand washing station, nurse station, storage space for stretchers, supplies and monitors / equipment and gas, suction outlets and ventilator.
- **Staff room** – Men and women change dress from street cloth to OT attire; lockers and lavatory are essential; rest room etc. are desirable.
- **Sanitary facility for staff**- One washbasin and one western closet (WC) should be provided for 8-10 persons. Showers and their number is a matter of local decision. Inclusion of toilet facilities in changing rooms is not acceptable; they should be located in an adjacent space.
- **Offices** – for staff nurse and anaesthesia staff- The office should allow access to both unrestricted and semi-restricted areas as frequent communication with public is needed.
- **Rest rooms**- Pleasant and quiet rest for staff should be arranged either as one large room for all grades of staff or as separate rooms; both have merits. Comfortable chairs, one writing table, a book case etc., may be arranged.
- **Laboratory** – Small laboratory with refrigerator for pathologist to be arranged.
- **Seminar room**- Since staff cannot leave an OT easily, it is better to have a seminar room within the OT. Intra-departmental discussions, teaching and training sessions for staff (with audio-visual aids) may be conducted here.
- **Store room**-
 - These rooms should have steel cabinets & separate for specific items.
 - These cabinets should be spacious enough to accommodate various items such as sterile linens, sterile trays, sterile drums with sponges, dressings, cotton swabs, bandages, sterile rubber sheets, catheters, sutures, syringes, infusion & transfusion sets.

- A separate store room is needed for unsterile, clean articles such as linen, OT dress items, stainless steel equipment's
- The cabinets of store rooms should be made up of such materials which can be cleaned & washed without being damaged.
- There should be bigger store room for equipment's such as monitoring devices, suction apparatus, O₂ cylinder, trolleys, wheel chairs, stretchers, defibrillator, pace makers, bed urinals, OT slippers etc.
- **Scrub room-** This is planned to be built within the restricted area. This should be spacious. Sometime s two or three operation theatres can share one scrub facility. Scrub area should have deep & wide sink to avoid splash of water on the surrounding area. There should be facilities for running water. The taps should be such that they could be opened or closed by foot pedals or they have long handles that can be operate by elbows.
- **Waiting room** – Patient's attendants or family member's waiting room with attached toilet & drinking water facility.
- **Visitor's gallery** - Gallery from where the visitors can see the operative procedures being performed.
- **Utility Room** - This is also called clean up room where instruments, articles, gloves are received following surgical procedures. Here all these are cleaned, washed & wiped dry & then wrapped, stored & sent for sterilization.
- **Disposal or sluice room** - This room receives all soiled linen & dressing. Here all disposable items are separated according to biochemical waste management code & then packed in leak proof containers. The soiled linens are sent to the laundry. The OT set up should be such that it should be possible to dispose off soiled items without taking them through sub-sterile or sterile areas.

OPRETING ROOM

NUMBER & SIZE

- The number & size can be as per the requirement.
- The standard OT should be rectangular or square in shape and similar in design.
- It should be 20×20×10 feet in size so it provides a floor space of 400 Sq. feet approx.

- It should be spacious enough to allow free movement of personnel, trolleys, stretchers, wheel chairs, monitoring devices, portable x-ray machines etc.
- Cardiac or neurosurgery OT should be bigger in size of 20x30x10 feet with 600 sq. feet floor space.
- OT for endoscopy or minor surgery can be of small size of 18x18x10 feet with a floor space of 324 sq. feet.

DOORS

- Main door to the OT suite has to be of adequate width (1.2 to 1.5 m).
- The doors of each OT should be spring loaded flap type, but sliding doors are preferred as no air currents are generated.
- All fittings in OT should be flush type and made of steel.

SURFACE / FLOORING

- The surface / flooring must be slip resistant, strong with minimum joints.
- It should be easily washable, free as possible joints, seams so that they do not permits adherence of bacteria or dust particles containing microorganism.
- It should be able to withstand repeated washing with germicidal agents.
- It should be able to absorb sound.
- The colour of the flooring should be such that if a needle is dropped on the floor it is visible.

WALLS

- The walls, ceiling & floor surfaces should be made of hard fire resistant, smooth, non-porous material.
- It should be light in colour (light blue or green) and washable paint should be ideal.
- Colour of paint should allow reflection of light and yet soothing to eyes.
- Adequate electric points should be available on the wall at < 1.5 m height from the floor.

SCRUB STATION

- For sterile hand wash
- Types: Stainless/ Granite/Marble/ Cement

- Gradient of basin forward and downwards
- Gradient towards outlet
- Hot water mixing: automatic or manual
- Non splashing taps; possible to change direction
- Elbow/ Foot or Infra-Red operated taps
- Soap dispenser: Manual/automatic
- to be planned for at least for 2-3 persons in each OT.

VENTILATION, TEMPERATURE & HUMIDITY

- The OT Require Efficient ventilation that will control temperature and humidity in OT and dilute the contamination by microorganisms and anaesthetic agents.
- The ventilation in OT should follow the principles that the direction of airflow should take place from the sterile to the clean zone & from the clean zone to the less clean areas.
- There should be no open window in the OT and sliding doors of the OT should be open from the OT towards the sub sterile zone.
- There should be no interchange air movement between one OT and another.
- There should be no circulating fan or cross ventilation to prevent airborne contamination of the surgical wound.
- An effective ventilation system is necessary to exchange the air. This air is filtered with an efficient air filtering system.
- There are two types of air conditioning systems : recirculating and non recirculating.
- The circulating system takes some or all of the air, adjusts the temperature and circulates air back to the room.
- Non-recirculating systems heat / cool the air as desired and convey it into the operating room with ideally 20-air exchange per hour. Air is then exhausted to outside. Anaesthetic agents in the OT air are also automatically removed. These are thus ideal but are expensive.
- The broad recommendations include:
 - 20-30 air exchanges / hour for recirculated air

- Only up to 80% recirculation of air to prevent build-up of anaesthetic and other gases
- Ultraclean laminar airflow – the filtered air delivery must be 90% efficient in removing particles more than 0.5µm.
- Positive air pressure system in OT: It should ensure a positive pressure of 5 cm H₂O from ceiling of OT downwards and outwards, to push out air from OT.
- Relative humidity of 50-60% to be maintained
- Temperature between 18o-24oC. Temperature should not be adjusted for the comfort of OT personnel but for the requirement of patient, especially in paediatric, geriatric, burns, neonatal cases etc.

LIGHTING

- The general room illumination the OT is provided by fluorescent lamps (surface ceiling mounted) to produce even illumination of at least 500 Lux at working height, with minimal glare are preferred.
- This contrast should be maintained in corridors and scrub areas, as well as in the room itself, so that the surgeon becomes accustomed to the light before entering the sterile field.
- To minimize eye fatigue, the ratio of intensity of general room lighting to that at the surgical site should not exceed 1:5, preferably 1:3.
- Colour and hue of the lights also should be consistent.
- The surgical or operation light should be an overhead ceiling mounted unit.
- The overhead operating light must:
 - Overhead light should be near daylight in colour and shadow less and give 25000-125000 Lux of light (50000 to 100000 Lux at the centre and at least 15000 Lux at the periphery).
 - Give contrast to the depth and relationship of all anatomic structures. The light may be equipped with an intensity control mechanism. The surgeon may ask for more light when needed therefore a reserve light should be available (e.g. a mobile operation light).

- Provide the diameter light pattern of a focus appropriate for the size of the incision. These are adjusted with controls mounted on the light fixture.
- Be freely adjustable to any position or angle. Most overhead operating lights are ceiling mounted on mobile fixtures. It can be positioned so that light is directed into a single incision or two concurrent operative sites.
- Be spark-proof where anaesthetic gases are used.
- Produce minimum heat to prevent injury to exposed tissues, to ensure the comfort of the sterile team, and to minimize airborne microorganisms.
- Be easily cleaned. Tracks recessed within the ceiling virtually eliminate dust accumulation. The radiant heat produced by the light should raise the tissue temperature not more than 2°C.

EMERGENCY SIGNAL

- The OT management committee should consider installing an alarm system in case of a life-threatening emergency.
- The activation of this alarm will save valuable time to mobilize the emergency designate staff and direct them to the site of the emergency.

OTHER FACILITIES OF OR

- **Electric generator** - OT department should have electric generator to ensure uninterrupted electricity supply in case of electricity failure.
- **O₂ supply & suction system** - OT should have piped in O₂ supply through central O₂ supply system from a central source. There should also be central suctioning through piped in suction arrangement.
- **Music system** - Operation theatre can also have a soft music system. Music creates a pleasant environment for patients & staff. Music also provide diversion of the patient's mind. Music can provide relaxation to pts who undergo surgery under local, regional or spinal anaesthesia. It also provide a soothing atmosphere & decreases tension & fatigue of the OT staff.

- **Computer terminal** - The modern OT construction also provide a surface area for the computer terminal in each OT. These computers are being used to record & maintain patient information

Sterilization in operation theatre

The methods of sterilization in common use are:

Sterilization in Operation Theatre

Bacterial endospores (e.g., clostridia and bacillus) are the most resistant of all living organisms because of their capacity to withstand external destructive agents. Although the physical or chemical process by which all pathogenic and non-pathogenic microorganisms (including endospores) are destroyed is not absolute, supplies and equipment are considered sterile when all parameters have been met during a sterilization process.

Reliability Parameters for Sterilization

Two types of parameters are considered for the reliability of sterilizing methods: product-associated parameters and process-associated parameters.

Product-Associated Parameters

- **Bio burden.** The degree of contamination with microorganisms and organic debris
- **Bio resistance.** Factors such as heat and/or moisture sensitivities and product stability
- **Bio state.** The nutritional, physical, and/or reproductive phase of microorganisms
- **Bio shielding.** Characteristics of the packaging materials
- **Density.** Factors affecting penetration and evacuation of the agent

Process-Associated Parameters

- Temperature
- Humidity/moisture/hydration
- Time
- Purity of the agent and air, and the residual effects or residues
- Saturation/penetration
- Capacity of the sterilizer and the position of items within the chamber

Methods of Sterilization

Reliable sterilization depends on the contact of the sterilizing agent with all surfaces of the item to be sterilized. Selection of the agent used to achieve sterility depends primarily on the nature of the item to be sterilized. The time required to kill endospores in the available equipment then becomes critical. Sterilization processes are either physical or chemical, and each method has its advantages and disadvantages. The following are available sterilizing agents (sterilants):

1. Thermal (physical)

- ✓ Steam under pressure/moist heat
- ✓ Hot air/dry heat

2. Chemical

- ✓ Ethylene oxide gas
- ✓ Formaldehyde gas and solution
- ✓ Hydrogen peroxide plasma/vapor
- ✓ d Ozone gas
- ✓ Acetic acid solution
- ✓ Glutaraldehyde solution
- ✓ Peracetic acid 0.2% solution
- ✓ Hypochlorous acid (electrochemical conversion process)

3. Radiation (physical)

- ✓ Microwave (nonionizing)
- ✓ b.X-ray (ionizing)

Sterilization Cycle

The time required to achieve sterilization is referred to as the process cycle, which includes the following:

- ✓ •Heat up and/or penetration of the agent
- ✓ •Kill time (i.e., exposure to the agent)
- ✓ •Safety factor for bioburden
- ✓ •Evacuation or dissipation of the agent

Chemical Sterilizing Agents

The chemical agents used to sterilize heat sensitive items can be toxic or can vaporize to emit noxious fumes that are irritating to the eyes and nasal passages, even at low levels of exposure.

Ethylene Oxide

EO is used in a gaseous form for sterilization and is known mutagen and carcinogen. The residual products can be toxic if there is direct contact with the skin or the gas is inhaled. Exposure can cause dizziness, nausea, and vomiting. Ethylene glycol and ethylene chlorohydrin are by products of a reaction with moisture, such as on the hands. All porous items sterilized with EO should be aerated to dissipate the gas.

Formaldehyde

Formaldehyde may be used in a gaseous or liquid form. The vapors are toxic to the respiratory tract. Formaldehyde is a potent allergen, mutagen, and carcinogen, and it can cause liver toxicity.

Glutaraldehyde

Glutaraldehyde is the less toxic of the three sterilizing agents, but the fumes from the liquid form may be irritating to the eyes, nose, and throat. Contact dermatitis and hives have been reported. Glutaraldehyde should be used only in a closed container and in a well-ventilated area.

Thermal Sterilization

Heat is a dependable physical agent for the destruction of all forms of microbial life, including endospores. It may be used moist or dry. The most reliable and commonly used method of sterilization is steam under pressure.

Steam under Pressure (Moist Heat Sterilization)

Heat destroys microorganisms, and this process is hastened by the addition of moisture. Steam in itself is inadequate for sterilization. Pressure that is greater than atmospheric pressure is necessary to increase the temperature of steam for the thermal destruction of microbial life. Moist heat in the form of steam under pressure causes the denaturation and coagulation of the protein or enzyme protein within the cells. Direct saturated steam contact is the basis of steam sterilization process. For a specified time and at a required temperature, the steam must penetrate every fibre and reach every surface of the items to be sterilized. When steam enters the sterilizer chamber under pressure, it condenses on contact with cold items. This condensation liberates heat, simultaneously heating and wetting all items in the load and thereby providing the two requisites:

Moisture and heat. This sterilization process is spoken of in terms of degrees of temperature and time of exposure-not in terms of pounds of pressure. Pressure increases the boiling temperature. Exposure time depends on the size and contents of the load and the temperature within the sterilizer. At the end of the cycle, reevaporation of water condensate must effectively dry contents of the load to maintain sterility; the water is dried from the sterilized pack or item. The vegetative forms of most microorganisms are killed in a few minutes at temperatures ranging from 130° F to 150° F (54° C-65°C); however, bacterial endospores will withstand a temperature of 240° F (115° C) for more than 3 hours. No living thing can survive direct exposure to saturated steam at 250° F (121°C) for longer than 15 minutes. As the temperature of the steam is increased, the time of exposure may be decreased. A minimum temperature-time relationship is maintained throughout all portions of the load to accomplish effective sterilization.

Advantages of Steam Sterilization

- i. sterilization is the easiest, safest, and surest method of onsite sterilization. Heat-and moisture-stable items that can be steam sterilized without damage should be processed with this method.
- ii. Steam is the fastest method; its total time cycle is the shortest.
- iii. Steam is the least expensive and most easily supplied agent. It is piped in from the facility's boiler room. An automatic, electrically powered steam generator can be mounted beneath the sterilizer for emergency standby when steam pressure is low.
- iv. Most sterilizers have automatic controls and recording devices that eliminate the human factor from the sterilization process as much as possible when operated and cared for according to the recommendations of the manufacturer.
- V. Steam leaves no harmful residue. Many items such as stainless steel instruments withstand repeated processing without damage.

Disadvantages of Steam Sterilization

- i. Precautions must be used in preparing and packaging items, loading and operating the sterilizer, and drying the load.
- ii. Items need to be clean, free of grease and oil, and not sensitive to heat.
- iii. Steam must have direct contact with all areas of an item. It must be able to penetrate packaging material, but the material must be able to maintain sterility.

iv. The timing of the cycle is adjusted for differences in materials and sizes of loads; these variables are subject to human error.

V. Steam may not be pure. Steam purity refers to the amount of solid, liquid, or vapor contamination in steam. Impurities can cause wet or stained packs and stained instruments,

Dry Heat Sterilization

and talc, which steam and ethylene oxide gas cannot penetrate. The destruction of microbial life by dry heat is a physical oxidation or slow burning process that involves coagulating the protein in cells. Higher temperatures are required in the absence of moisture because the microorganisms are destroyed through a very slow process of heat absorption by conduction. Dry heat sterilizers are not commonly found in the OR processing areas. Dry heat is commonly used in dental suites,

Advantages of Dry Heat Sterilization

- i. Hot air penetrates certain substances that cannot be sterilized by steam sterilization or another method.
- ii. Dry heat is a protective method of sterilizing some delicate, sharp, or cutting-edge instruments. Steam may erode or corrode cutting edges.

Disadvantages of Dry Heat Sterilization

- i. A long exposure period is required, because hot air penetrates slowly and possibly unevenly.
- ii. The time and temperature required will vary for different substances.
- iii. Overexposure may ruin some substances.

Safety practices in Operation Theatre

Surgical care is a fundamental part of healthcare around the globe. Ensuring surgical safety in operation theatres is the need of the hour. A safe operating theatre is achieved only through careful planning, maintenance and periodic checks, as well as proper ongoing training for staff members of the structure and its fixtures, but also the management and behavior of healthcare workers entering the sterilizer.

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1. **Autoclaving** · Autoclaving should be the main form of sterilization at the district hospital. · Before sterilizing medical items, they must first be disinfected and vigorously cleaned to remove

all organic material. Proper disinfection decreases the risk for the person who will be cleaning the instruments.

· Sterilization of all surgical instruments and supplies is crucial in preventing HIV transmission. All viruses, including HIV, are inactivated by steam sterilization (autoclaving) for 20 minutes at 121°C–132°C or for 30 minutes if the instruments are in wrapped packs. · Appropriate indicators must be used each time to show that sterilization has been accomplished. At the end of the procedure, the outsides of the packs of instruments should not have wet spots, which may indicate that sterilization has not occurred.

2. **Dry heat** · If items cannot be autoclaved, they can be sterilized by dry heat for 1–2 hours at 170°C. Instruments must be clean and free of grease or oil. · Sterilizing by hot air is a poor alternative to autoclaving since it is suitable only for metal instruments and a few natural suture materials. · Boiling instruments is now regarded as an unreliable means of sterilization and is not recommended as a routine in hospital practice.

3. **Antiseptics** · Sharp instruments, other delicate equipment and certain catheters and tubes can be sterilized by exposure to formaldehyde, glutaral (glutaraldehyde) or chlorhexidine. · If you are using formaldehyde, carefully clean the equipment and then expose it to vapour from paraformaldehyde tablets in a closed container for 48 hours. Glutaral is a disinfectant that is extremely effective against bacteria, fungi and a wide range of viruses. Always follow the manufacturer's instructions for use.

Modern Issues of Patient Safety

Medical errors are inescapable in healthcare profession. The identification of causes and implementation of safety plans to reduce these errors will help to establish an effective patient safety in operation theatres. The causes advocated by WHO are enlisted below:

- ✓ Lack of continuous training and education
- ✓ Past tolerance of unsafe practice
- ✓ Lack of regulations /rules
- ✓ Gaps in communication among different healthcare providers
- ✓ Gaps in communications between healthcare providers and patients
- ✓ Unstable /unreliable systems
- ✓ Fear of admission of guilt / wrong doing

- ✓ Human factors

PACKING OF SURGICAL INSTRUMENTS

Packing implies the process of wrapping up of product into a case or container, for the purpose of protection.

TYPES OF PACKING MATERIALS AVAILABLE FOR STERILIZING

Packaging materials include wrapped perforated instrument cassettes, peel pouches of plastic or paper, and sterilization wraps (which can be either woven or unwoven). Packaging materials should be designed for the type of sterilization process being used and should be appropriate for the items being sterilized.

Packing of surgical instruments

To be effective the sterilizing agent must come into direct contact with all surfaces of every instrument. Therefore instruments must be packaged (individually or in sets) in such a way to allow adequate exposure to the sterility, prevent air from being trapped and moisture from being retained during the sterilization process, and ensure sterile transfer to the sterile field. The majority of surgical instruments are made of stainless steel and can be sterilized by steam under pressure. Effective steam sterilization involves the direct contact of all surfaces with steam and the revaporization of water condensate to produce a dry, sterile instrument.

Instrument Packaging

For sterilizing and transporting, instruments are put in a closed container or wrapped individually. Instruments placed in open trays are wrapped as sets. Instruments may be sterilized unwrapped in a high-speed steam sterilizer immediately before use (not recommended), may be prepared in advance as for a case cart, or may be retained in sterile core storage until needed.

Packaging Considerations

The packaging materials for all methods of sterilization should do the following:

- Permit penetration of the sterilizing agent to achieve sterilization of all items in the package.
- Allow the release of the sterilizing agent at the end of the exposure period and allow adequate drying or aerating.

- Withstand the physical conditions of the sterilizing process.
- Maintain integrity of the package at varying atmospheric and humidity levels. In dry climates or at high altitudes, some packaging materials are susceptible to rupture during sterilization or dry out and crack in storage.
- Provide an impermeable barrier to microorganisms, dust particles, and moisture after sterilization. Items must remain sterile from the time they are removed from the sterilizer until they are used.
- Cover items completely and easily and fasten securely with adhesive strip or a heat seal that cannot be resealed after opening. Seal integrity should be tamperproof. A margin of at least 1 inch (2.5 cm) is considered a standard for safety on all sealed packages. Scissors should not be used to cut off the end of a package. Contents cannot be drawn out over this cut end because the contents would be contaminated by the edge of the packaging material. For the same reason, packages are never torn open below a seal.
- Resist tears and punctures in handling. If accidental tears and holes do occur, they must be visible.
- Identification of the contents and the chemical indicator evidence of exposure to a sterilizing agent should be placed on a label that is visible on the outside of the package.
- Be free of toxic ingredients and nonfast dyes.
- Be lint-free or low-linting.
- Protect the contents from physical damage.
- Permit easy removal of the contents with transfer to the sterile field without contamination or delamination (separation into layers).
- Be economical.

Packages should be wrapped far enough away from sterile storage areas so that mixing sterile and non-sterile packages is not possible. Cabinets that contain non-sterile items should be labelled conspicuously. A procedure for sending items to and receiving items from the sterilizer should be set up so that sterile and non-sterile packages can never be confused en route. This procedure must be understood by everyone.

Packages may be wrapped with nonwoven or woven materials sequentially in two layers. Double-thickness wrapping in one layer without using the sequential method is also acceptable.

The wrappers can be stitched around the edges or bonded/fused together. Aseptic presentation to the sterile field is the prime consideration. Each facility should determine which method and material is best suited to the clinical environment.

Video link on packaging of surgical instruments

<https://www.youtube.com/watch?v=FF5D6flcra0>.

Theatre layout

The overall floor plan of the surgical suite is divided into three areas or zones, which are directly or indirectly involved with the operative procedure equipment, supplies, and personnel.

For descriptive purposes the zones represent the type of activities, dress code, or restriction for that zone. Each person working within the suite must abide by policies related to the zones.

The Three-Zone Concept

Types and Use of Sterilization Packaging Materials

Sterilization Method	Packaging Material Requirements	Acceptable Materials
Steam autoclave	• Should allow steam to penetrate	• Paper • Plastic • Cloth • Paper/plastic peel packages • Wrapped perforated cassettes
Dry heat	• Should not insulate items from heat • Should not be destroyed by temperature used	• Paper bags • Aluminum foil • Polyfilm plastic tubing • Wrapped perforated cassettes
Unsaturated chemical vapor	• Vapors should be allowed to precipitate on contents • Vapors should not react with packaging	• Wrapped perforated cassettes • Paper

Sterilization Method	Packaging Material Requirements	Acceptable Materials
	material Plastics should not contact sides of sterilizer	Paper/plastic peel packages

Considerations to prevent packaging tears

- ✓ The packaging should be snug to the instrument set.
- ✓ Reduce the amount of instruments in a set. Heavy instrument sets are a common cause of packaging tears.
- ✓ Use the correct packaging weight. Heavy instrument sets require thicker packaging.

Before placing packaged instruments in the sterilizer, at a minimum, include the following information on the label:

- ✓ Sterilizer used
- ✓ Cycle or load number
- ✓ Date of sterilization
- ✓ Expiration date, if applicable

PREVENTION OF NOSOCOMIAL INFECTIONS 2

Laminar flow system and its functions

A **laminar flow cabinet** or **tissue culture hood** is a carefully enclosed bench designed to prevent contamination of semiconductor wafers, biological samples, or any particle sensitive materials.

Air is drawn through a HEPA filter and blown in a very smooth, laminar flow towards the user. Due to the direction of air flow, the sample is protected from the user but the user is not protected from the sample. The cabinet is usually made of stainless steel with no gaps or joints where spores might collect.

Laminar flow cabinets may have a UV-C germicidal lamp to sterilize the interior and contents before usage to prevent contamination of the experiment.

Germicidal lamps are usually kept on for fifteen minutes to sterilize the interior before the cabinet is used.

The light must be switched off when the cabinet is being used, to limit exposure to skin and eyes as stray ultraviolet light emissions can cause cancer and cataracts

Mechanisms and Functions of laminar flow system

- The contaminated air enters through the flow hood and gets to the filters.
- Laminar flow hoods use two types of filters; the pre-filter, and HEPA filter.
- The air is channelized through a secondary blower by the pre-filter. Thus, it is a safe process.
- After that, it is passed through the HEPA filter.
- The air is cleaned through the process, leaving the samples of the laboratory safe from pollutants.

FUNCTIONS

The two major purposes of laminar flow hoods are filtering the air and protecting scientific substances. Apart from laboratories, these features come in use in various industries including,

- Research regarding plants.
- Cleaning industries.
- Recovery of data.

- Clinical applications.
- Stem cell therapy.
- Designing and manufacturing projection lens.
- Tissue culture.

Types of Laminar Workstations

Laminar flow workstations are available in two configurations:

- **Horizontal**

The air is direct from above that changes direction and then processes it throughout the workstation in a horizontal manner. The constant air supply keeps the material placed inside protected.

- **Vertical**

This type of configuration works in the same manner as a horizontal laminar flow workstation. The only difference is that the air reaches a vertical mode and then escapes through the base's holes. This configuration provides sufficient protection to operators.

- **Exponential laminar flow**

Advantages of Using Laminar Workstations

The benefits of a laminar airflow workstation depend on the application they boast designs and functionality for a specialized setting. However, some common positions that apply for all workstations are as under:

- **Environmental Friendly Features**

Perhaps the most advantageous feature of a laminar flow workstation is the eco-friendly functions it possesses. The workstations recycle air in the room and clear the contaminations present at the same time. In short, no harmful air harms the environment outside the laboratory, and filters can be disposed of in landfills without any damage to the surroundings.

- **Portability**

Another advantageous feature of these workstations is that they are portable. There are no other fixtures present beside power cords, making them accessible and easy to take wherever you want. In the absence of an HVAC system or vents, these devices are smooth to move from one place to another and use accordingly.

- **Safety Functions**

Laminar flow workstations employ the use of monitoring systems to check the saturation of gas and carbon. There are alarms present with the monitor that can notify of any danger level as per the safety standards.

- **Economical**

Despite their usefulness and functionality, laminar flow workstations are relatively economical devices. The only maintenance cost required after installation is the replacement of filters annually. There is minimal cost associated with their transportation and assembly.

PATIENT IDENTIFICATION PROCESS IN OT

Patient identifier options include:

1. Name.
2. Assigned identification number (e.g., medical record number)
3. Date of birth.
4. Phone number.
5. Social security number.

6. Address.

7. Photo.

1. ****Patient Identification and Consent****:

- Verify the patient's identity using their full name, date of birth, and medical record number.
- Confirm the correct patient, procedure, surgical site, and side with the patient and mark the site as needed.
- Obtain informed consent from the patient after explaining the procedure, risks, benefits, and alternatives. Ensure the patient understands and signs the consent form.

2. ****Medical History and Assessment****:

- Review the patient's medical history, including past surgeries, allergies, current medications, and chronic conditions.
- Perform a comprehensive physical examination to assess the patient's overall health status, including vital signs, cardiovascular and respiratory function, and neurological assessment.
- Assess the patient's nutritional status, hydration level, and any preexisting medical conditions that may impact the surgery.

3. ****Preoperative Testing and Evaluation****:

- Order and review preoperative tests, such as blood work, imaging studies, electrocardiogram (ECG), and other diagnostic tests based on the patient's medical history and the planned procedure.
- Arrange for consultations with specialists, such as an anesthesiologist or cardiologist, if needed to evaluate the patient's fitness for surgery.

4. ****Medication Management****:

- Review the patient's current medications, including prescription, over-the-counter, and herbal supplements.
- Adjust medications as necessary before surgery, particularly anticoagulants or medications that may interact with anesthesia.

- Provide instructions on which medications to take or withhold on the day of surgery.

5. ****Preoperative Instructions and Preparation****:

- Provide detailed preoperative instructions to the patient, including fasting guidelines, showering with antiseptic soap, and dressing in clean, loose-fitting clothing.
- Advise the patient on specific dietary restrictions, smoking cessation, and alcohol consumption before surgery.
- Clarify arrival time, parking instructions, and any necessary paperwork or identification required for admission.

6. ****Psychological and Emotional Support****:

- Address the patient's emotional needs and concerns about the surgery, including anxiety, fear, or stress.
- Offer emotional support, answer questions, and provide resources for coping strategies or counseling if needed.
- Ensure the patient feels comfortable, informed, and supported throughout the preoperative process.

7. ****Anesthesia Evaluation and Planning****:

- Coordinate a preoperative anesthesia evaluation to assess the patient's medical history, airway, cardiovascular status, and anesthesia risks.
- Collaborate with the anesthesiologist to determine the appropriate anesthesia plan, including type of anesthesia, monitoring devices, and postoperative pain management.

8. ****Surgical Site Preparation and Infection Control****:

- Ensure the surgical site is properly prepared according to surgical protocols, including hair removal, skin cleansing, and draping.
- Follow strict infection control measures, such as hand hygiene, sterile technique, and proper use of surgical attire, to reduce the risk of surgical site infections.

By following these detailed steps and individualizing care based on the patient's needs and the planned procedure, healthcare providers can help ensure a safe and successful surgical experience for the patient.

Lesson 1

PATIENT POSITIONING.

Surgical positioning is the **practice of placing a patient in a particular physical position during surgery**. The goal in selecting and adjusting a particular surgical position is to maintain the patient's safety while allowing access to the surgical site.

REASONS FOR PATIENT POSITIONING DURING SURGERY

- The goals of positioning the surgical patient are
- ensuring patient comfort and dignity;
- maintaining homeostasis;
- protecting anatomical structures and avoiding complications and injuries;
- promoting access to the surgery site;
- promoting access for the administration of IV fluids and anesthetic agents;
- promoting access of OR surgical equipment.

TYPES OF PATIENT POSITIONS

Supine position

The most common surgical position. The patient lies with back flat on operating room bed.(frontal organ surgeries i.e stomach, lung)

Trendelenburg position

Same as supine position but the upper torso is lowered. (lung problems)

Reverse Trendelenburg position

Same as supine but upper torso is raised and legs are lowered.(gastrointestinal problems)

Fracture Table Position YG 0

For hip fracture surgery. Upper torso is in supine position with unaffected leg raised. Affected leg is extended with no lower support. The leg is strapped at the ankle and there is padding in the groin to keep pressure on the leg and hip.(hip surgery)

Lithotomy position

Used for gynecological, anal, and urological procedures. Upper torso is placed in the supine position, legs are raised and secured, arms are extended.(vaginal examination and child birth)

Fowler's position

Begins with patient in supine position. Upper torso is slowly raised to a 90 degree position.(cardiac, respiratory or neurological problems)

Semi-Fowlers position

Lower torso is in supine position and the upper torso is bent at a nearly 85 degree position. The patient's head is secured by a restraint.

Prone position

Patient lies with stomach on the bed. Abdomen can be raised off the bed. Researchers observed that patients can benefit from prone positioning.(neurosurgery, neck and spine surgery)

Jackknife position

Also called the Kraske position.^[2] Patient's abdomen lies flat on the bed. The bed is scissored so the hip is lifted and the legs and head are low.anal surgery,rectum,coccyx, vertain back surgeries)

Knee-chest position

Similar to the jackknife except the legs are bent at the knee at a 90 degree angle.(sigmoidoscopy)

Lateral position

Also called the side-lying position, it is like the jackknife except the patient is on his or her side. Other similar positions are Lateral chest and Lateral kidney. (helps relieving pressure sacrum and heels)

Lloyd-Davies position

It is a medical term referring to a common position for surgical procedures involving the pelvis and lower abdomen. The majority of colorectal and pelvic surgery is conducted with the patient in the Lloyd-Davis position.

Kidney position

The kidney position is much like the lateral position except the patient's abdomen is placed over a lift in the operating table that bends the body to allow access to the retroperitoneal space. A kidney rest is placed under the patient at the location of the lift.(abdominal, peritoneal problems)

Sims' position

The Sims' position is a variation of the left lateral position. The patient is usually awake and helps with the positioning. The patient will roll to his or her left side. Keeping the left leg straight, the patient will slide the left hip back and bend the right leg. This position allows access to the anus. (anal surgery)

PREPPING AND DRAPING THE PATIENT

The procedure of prepping the patient involves several steps to ensure the patient's surgical site is adequately cleaned and prepared for the procedure. Here is a general outline of the steps involved in prepping the patient:

1. Review the patient's medical history and confirm the procedure: Review the patient's medical records to ensure that the correct procedure is being performed. Confirm any specific considerations or contraindications related to the patient's medical history.
2. Explain the prepping process to the patient: Communicate with the patient and explain the purpose and steps involved in the prepping process. Address any concerns or questions they may have to ensure their understanding and cooperation.

3. Prepare the environment: Ensure that the prepping area is clean, well-lit, and equipped with the necessary supplies. Remove any unnecessary equipment or clutter that may hinder the prepping process.
4. Hand hygiene: Perform proper hand hygiene by washing your hands thoroughly with soap and water or using an alcohol-based hand sanitizer.
5. Positioning the patient: Position the patient appropriately for the procedure, ensuring their comfort and accessibility to the prepping area. Provide any necessary support or padding to maintain their comfort during the process.
6. Privacy and draping: Respect the patient's privacy by ensuring they are appropriately covered during the prepping process. Use curtains or screens to create a private area if necessary. Use sterile drapes or barriers to create a sterile field around the prepping area, leaving only the surgical site exposed.
7. Skin preparation: Cleanse the patient's skin at the surgical site using an appropriate antiseptic solution recommended for the specific procedure. Follow the recommended technique, typically starting from the center and working outward in a circular motion. Allow the antiseptic solution to dry completely before proceeding.
8. Applying adhesive drapes or incise towels: If needed, apply adhesive drapes or incise towels to create a sterile barrier around the surgical site. Ensure that they are properly positioned and adhere securely to the patient's skin.
9. Documentation: Document the prepping process in the patient's medical record. Include details such as the antiseptic solution used, any complications or patient reactions, and the names of the healthcare professionals involved.
10. Maintain a sterile field: Throughout the prepping process, ensure that the sterile field is maintained. Avoid touching non-sterile surfaces or items once gloved and adhere to aseptic techniques.

11. Communicate with the surgical team: Maintain clear communication with the surgical team during the prepping process. Coordinate movements and ensure that everyone is aware of the sterile field boundaries to avoid accidental contamination.

It's important to note that the specific steps and techniques involved in prepping the patient may vary depending on the type of procedure, the surgical site, and the protocols followed by the healthcare facility. It's crucial to adhere to the guidelines provided by your healthcare facility and the specific procedure's protocol to ensure proper prepping and minimize the risk of infection.

DRAPING THE PATIENT

Draping the patient is a common practice in healthcare settings, particularly during medical procedures or examinations. It involves covering the patient's body with a sterile or clean drape to maintain their privacy, ensure infection control, and create a sterile field for the procedure. Draping is essential for maintaining patient dignity and preventing the spread of infection.

During surgical procedures, draping is performed to create a sterile barrier between the patient's surgical site and the surrounding non-sterile environment. The drapes used are typically made of disposable materials and are designed to be impermeable to liquids and microorganisms. The surgical team carefully positions the drapes to cover the patient's body, leaving only the surgical site exposed.

In non-surgical settings, draping is also commonly used during physical examinations or other procedures to maintain patient privacy and modesty. These drapes may be made of disposable or reusable materials, depending on the healthcare facility's protocols and infection control measures.

When draping the patient, healthcare professionals should follow certain guidelines to ensure effectiveness and patient comfort:

1. Proper positioning: The patient should be positioned appropriately for the procedure, ensuring access to the area being examined or treated. The drapes should be placed to cover the areas that do not require exposure.

2. Maintaining sterility: In surgical settings, sterile drapes must be used to create a sterile field. Healthcare professionals should avoid touching non-sterile surfaces or equipment while draping the patient. If sterile integrity is compromised, the drapes should be replaced.
3. Communication and consent: Healthcare professionals should explain the purpose of draping to the patient and obtain their consent before proceeding. Patients should be given the opportunity to ask questions or express any concerns regarding the procedure or draping process.
4. Patient comfort and modesty: Healthcare professionals should ensure that the drapes are positioned securely but comfortably, taking into consideration the patient's comfort and modesty. The patient's body should be adequately covered, and efforts should be made to minimize exposure during the procedure.

In summary, draping the patient is a crucial practice in healthcare settings to protect patient privacy, maintain sterility, and prevent the spread of infection. Whether in surgical or non-surgical settings, proper draping techniques should be followed to ensure patient comfort, maintain sterility, and adhere to infection control protocols.

THE PROCEDURE OF DRAPING THE PATIENT

The procedure of covering a patient and surrounding areas with a sterile barrier to create and maintain a sterile field during a surgical procedure is called draping. The purpose of draping is **to eliminate the passage of microorganisms between nonsterile and sterile areas.**

The purpose of draping is to eliminate the passage of microorganisms between nonsterile and sterile areas. Draping materials may be disposable or nondisposable. Disposable drapes are generally paper or plastic or a combination and may or may not be absorbent. Nondisposable drapes are usually double-thickness muslin. Drapes, of course, must be sterile.

Since draping is very important in preparing a patient for surgery, it must be done correctly. The entire surgical team should be familiar with the draping procedure. The scrub must know the procedure perfectly and be ready to assist with it. During the draping procedure, the circulator should stand by to direct the scrub as necessary and to watch carefully for breaks in sterile technique.

(1) The first step in draping is the placing of a drape sheet from the foot to the knees. The scrub will select the sheet and hand one end to the surgeon across the operating table, supporting the folds, keeping it high, and holding it taut until it is opened, then drop it (open fingers and release sheet). The second drape sheet is handled in the same manner. This sheet is placed below the incision site with the edge of the sheet just below the incision site. This draping sheet provides extra thickness of material under the area from the Mayo tray to the incision where instruments and sponges are placed. It also closes some of the opening in the laparotomy sheet, if necessary.

(2) When disposable drapes are used, the towels usually have a removable strip with an adhesive on the folded edge. The third step in draping is placing the four sterile towels around the line of incision. The scrub unfolds first towel, passes the towel drape to the surgeon with the strip side facing the scrub, and then removes the adhesive strip. The surgeon places the towel within the scrubbed area on the near side of the line of incision, leaving only enough exposed skin for the incision. The second towel placed in the same way, except the towel is placed on the lower side (toward feet) of the line of incision. The third towel is passed the same way, except the towel is placed on the upper side (toward head) the line of incision. The last towel is passed to the surgeon with the adhesive strip facing the surgeon and is placed on the far side of the line of incision. The adhesive area holds the towel drapes in place.

NOTE: The only procedure changes that are made with nondisposable, muslin drapes (for example, hand towels) are as follow. The towels are cuffed by the scrub about 3 inches and the folded edge goes next to the line of incision. The first three towels are cuffed toward the scrub; the fourth towel is cuffed toward the surgeon. The towels are held in place by towel clips rather than by adhesive.

(3) Finally, the scrub will select the surgical drape (lap sheet). This lap sheet has a fenestration (opening) in the drape for the incision. The scrub places the opening directly over the skin area outlined by the drape towels and in the direction indicated for the foot or head of the table. The lap sheet will have an arrow or some other indication to identify the head or foot portion of the drape. Drop the folds over the sides of the table, then open it downward over the patient's feet and upward over the anesthetist screen.

Aseptic technique

Aseptic technique is **a collection of medical practices and procedures that helps protect patients from dangerous germs**. Bacteria, viruses, and microorganisms are everywhere, so using aseptic technique can help keep important equipment from being contaminated.

It includes the following;

- (1) Handle the drapes as little as possible.
- (2) Never reach across the operating table to drape the opposite side; go around the table.
- (3) Hold the drapes high enough to avoid touching nonsterile area but avoid touching the overhead light.
- (4) Hold the drape high until it is directly over the proper area, then drop (open fingers and release sheet) it down where it is to remain. **NEVER ADJUST ANY DRAPE**. If the drape is incorrectly placed, leave it in place and place another drape over it.
- (5) Protect the gloved hands by cuffing the end of the sheet over them. Do not let the gloved hand touch the skin of the patient.
- (6) In unfolding a sheet from the operative site toward the foot or head of the table, protect the gloved hand by enclosing it in the turned back cuff of the sheet.
- (7) If a drape becomes contaminated, discard it immediately.
- (8) If the end of a drape falls below waist level, do not handle it further. Drop it and use another drape.
- (9) If in doubt about sterility, discard the drape.
- (10) If a hole is found in a drape after it is laid down, cover the hole with another drape or discard the entire drape.

DRAPING MATERIALS

Draping materials used in medical procedures serve to create a sterile barrier between the patient's surgical site and the surrounding environment. They also help to maintain a clean and organized surgical field. Two commonly used draping materials are adhesive drapes and incise towels.

1. Adhesive drapes: Adhesive drapes are made of a thin, transparent plastic material with an adhesive backing. They are used to cover a wide area of the patient's body and provide an effective barrier against microbial contamination. Adhesive drapes are typically used to create a sterile field around the surgical site.

Key features and benefits of adhesive drapes include:

- Adhesive backing: The adhesive backing allows the drape to adhere securely to the patient's skin or surrounding drapes, maintaining its position throughout the procedure.
- Transparency: The transparent nature of adhesive drapes allows the surgical team to visually monitor the surgical site, ensuring proper visualization and access during the procedure.
- Fluid resistance: Adhesive drapes are designed to resist the penetration of liquids, such as blood or irrigation fluids, helping to keep the surgical field dry and clean.

2. Incise towels: Incise towels are sterile, absorbent towels with an adhesive edge. They are specifically designed to create a sterile barrier over the surgical incision site, preventing microbial contamination during the procedure. Incise towels are often used in procedures where a surgical incision will be made.

Key features and benefits of incise towels include:

- Adhesive edge: The adhesive edge of the incise towel allows it to be securely attached to the patient's skin, creating a barrier around the incision site. This helps to prevent the migration of bacteria from the surrounding non-sterile areas into the surgical site.

- Absorbency: Incise towels are made from absorbent materials that help to absorb any fluids or blood that may be generated during the procedure, keeping the incision site clean and dry.

- Fenestrated design: Incise towels often have a fenestration or hole in the center, which is precisely placed over the intended incision site. This allows the surgical team access to the incision area while maintaining a sterile environment.

Both adhesive drapes and incise towels are designed to maintain a sterile field during surgical procedures, reducing the risk of surgical site infections. They are available in various sizes to accommodate different surgical sites and procedures. It's important to follow manufacturer instructions and best practices when applying these draping materials to ensure their effectiveness in maintaining a sterile environment.

BASIC COMMUNICABLE AND NON COMMUNICABLE DISEASES^{2st} feb.

TUBERCULOSIS

Definition

A chronic and infectious mycobacterial disease important as a major cause of illness and death in many parts of the world.

Infectious agent.

Mycobacterium tuberculosis- human tubercle bacilli

(Commonest cause)

Mycobacterium bovis- cattle and man infection

Mycobacterium avium- infection in birds and man.

MODE OF TRANSMISSION

- Through aerosolized droplets mainly from persons with active ulcerative lesion of lung expelled during talking, sneezing, singing, or coughing directly.
- Transmission through contaminated fomites (clothes, personal articles) is rare.

- Ingestion of unpasteurized milk transmits bovine tuberculosis.
- Overcrowding and poor housing conditions favor the disease transmission.

Incubation period- 4-12 weeks

Period of communicability- as far as the bacilli is present in the sputum

Susceptibility and resistance- under 3 years old children, adolescents, young adults, the very old and the immunosuppressed are susceptible. Everyone who is non-infected or non-vaccinated can be infected.

- HIV is an important risk factor for the development of HIV associated tuberculosis by facilitating:
Reactivation, Progression of recent infection or Reinfection.

Clinical Manifestation of Pulmonary tuberculosis

- Persistent cough for 3 weeks or more
- Productive cough with or without blood-stained sputum
- Shortness of breath and chest pain
- Intermittent fevers, night sweats, loss of weight, loss of appetite, fatigue and malaise.

Nursing care

1. Educate the patient how and when to take the prescribed medication.
2. Tell the patient not to stop the medication unless he/she is told to do so.
3. Tell the patient to come to the health institution if he/she develops drug side effects.
4. Advise the patient on the importance of taking adequate and balanced diet and to eat what is available at home.

Prevention and control

1. Chemotherapy of cases

2. Chemoprophylaxis

INH (Isoniazid) for adults and children who have close contact with the source of infection

3. Immunization of infants with BCG

4. Educate patients with TB about the mode of disease transmission and how to dispose their sputum and cover their mouth while coughing, sneezing, etc.

5. Public health education about the modes of disease transmission and methods of control

- Improved standard of living
- Adequate nutrition
- Health housing
- Environmental sanitation
- Personal hygiene; etc.
- Active case finding and treatment

CHOLERA

Definition

An acute illness caused by an enterotoxin elaborated by **Vibrio cholerae**.

Reservoir- Humans

Mode of transmission- by ingestion of food or water directly or indirectly contaminated with feces or vomitus of infected person.

Incubation period- from a few hours to 5 days, usually 2-3 days.

Period of communicability- for the duration of the stool positive stage, usually only a few days after recovery. Antibiotics are used to shorten the period of communicability.

Susceptibility and resistance- Variable. Gastric achlorhydria (an absence of hydrochloric (HCl) acids in the stomach.) increases risk of illness. Breast-fed infants are protected..

Clinical Manifestation

- Abrupt painless watery diarrhea; the diarrhea looks like rice water. *f*
- In severe cases, several liters of liquid may be lost in few hours leading to shock.

- *f* Severely ill patients are cyanotic(**causing a bluish or purplish discoloration**), have sunken eyes and cheeks, scaphoid abdomen (poor skin turgor, and thready or absent pulse).
- *f* Loss of fluid continues for 1-7 days.

Diagnosis*f*

Based on clinical grounds *f* Culture (stool) confirmation

Treatment

1. Prompt replacement of fluids and electrolytes *f* Rapid IV infusions of large amounts *f* Isotonic saline solutions alternating with isotonic sodium bicarbonate or sodium lactate.
2. Antibiotics like tetracycline dramatically reduce the duration and volume of diarrhea resulting in early eradication of vibrio cholera.

Nursing care

1. Wear gown and glove.
2. Wash your hands.
3. Monitor output including stool output.
4. Protect the patient family by administering Tetracycline.
5. Health education.

Prevention and control

1. Case treatment
2. Safe disposal of human excreta and control of flies
3. Safe public water supply
4. Handwashing and sanitary handling of food
5. Control and management of contact cases

MALARIA

An acute infection of the blood caused by protozoa of the genus plasmodium.

Infectious agent. *f*

- Plasmodium falciparum/malignant tertian:
- Invades all ages of red blood cells. Red blood cell cycle is 48 hours *f*

- Plasmodium vivax/benign tertian: Invades reticulocytes only. Red blood cell cycle is 48 hours.
- Plasmodium ovale/tertian: Invades reticulocytes only. Red blood cell cycle is 48 hours. *f*
- Plasmodium Malariae/Quartan malaria: Invades reticulocytes only. Red blood cell cycle is 72 hours.

Predisposing factors are:

- Environment- physical environment for the propagation
- Patient source *f*
- Susceptible recipients
- Anopheles capable to transmit the parasite *f*
- Socio-economic factors like immigration, war, poverty, ignorance, agricultural irrigation farms, etc.

Mode of transmission-

- By the bite of an infective female anopheles mosquito, which sucks blood for egg maturation.
- Blood transfusion,
- hypodermic needles,
- organ transplantation and mother to fetus transmission is possible.
- Since there is no pre-erythrocytic (tissue) cycle, the incubation period is short

Incubation period- Varies with species *f*

- Plasmodium falciparum 7-14 days *f*
- Plasmodium vivax 8-14 days *f*
- Plasmodium ovale 8-14 days *f*
- Plasmodium malariae 7-30 days

Period of communicability- Mosquitoes are infective as long as infective gametocytes are present in the blood of patients. Once infected, mosquito remains infective for life.

Susceptibility and resistance- Susceptibility is universal except in some host-resistance factors

Non specific factors

- *f* Increased splenic clearance reaction
- *f* Hyperpyrexia- which is said to be schizontidal
- *f* Sick cell traits are resistant to plasmodium falciparum
- Because of passive immunity infants are resistant in early life.(A newborn baby acquires passive immunity from its mother through the placenta.)

Specific factors

This is a humoral and cell mediated immunity that is species and strain specific, and hard-won after repeated infection.

TRANSMISSION

1. Sporozoites inoculated when Anopheles mosquito takes a blood meal.

HUMAN HOST

2. Sporozoites infect liver cells. Multiply by schizogony. Note: some sporozoites of *P.vivax* and *P.ovale* become dormant hypnozoites in liver. Become active after several months.
3. Liver schizonts rupture. Merozoites enter red cells, become trophozoites. Multiply by schizogony.* * with *P. falciparum*, schizogony occurs in capillaries of body organs.
4. Schizonts rupture. Merozoites infect new red cells.
5. Some merozoites develop into male and female gametocytes.

MOSQUITO

6. gametocytes ingested by mosquito.
7. Male and female gametes fuse. Zygote oocyst in stomach wall.
8. Sporozoites form in oocyst.
9. Oocyst ruptures. Sporozoites reach salivary glands of mosquito.

Clinical manifestation;

Chills, rigor, fever, head ache, diarrhea, hallucinations, abdominal pain, aches, renal or respiratory symptoms, jaundice, etc.

Diagnosis

- Clinical manifestation and epidemiological grounds
- Blood film for hemoparasite
- White blood cell count
- Blood culture to rule out sepsis
- Chest X-ray to rule out pneumonia.

Treatment

1. Plasmodium vivax, ovale and sensitive plasmodium falciparum, Chloroquine or Fansidar
2. Chloroquine resistant falciparum and when sensitivity pattern is not known. Quinine or Fansida

Nursingcare

1. Advise patient to come back if the illness gets severe.
2. Advise on personal protection (bed nets, etc).
3. Reduce fever and maintain comfort.

Prevention and control

1. Chemoprophylaxis- for those who go to endemic areas but not for those who live in the endemic area (travelers and newcomers); for under-five children and pregnant mothers who have not enough immunity.
2. Vector control
 - Avoiding mosquito breeding sites
 - Residual DDT spray or other chemicals
 - Personal protection against mosquito bite (use of treated bed nets, etc.)
3. Chemotherapy of cases.

NON-communicable diseases (assignment)

Operation theatre ethics

Ethics –These are well-founded standards of right and wrong that prescribe what humans ought to do

Beneficence: Beneficence refers to the duty of healthcare professionals to act in the best interest of the patient. In the context of the operation theatre, this principle entails providing the highest quality of care and promoting the well-being of the patient. Healthcare professionals should strive to maximize benefits and minimize harm during surgical procedures.

2. **Non-maleficence:** Non-maleficence emphasizes the principle of "do no harm." In the operation theatre, this means taking all necessary precautions to prevent harm to the patient. Healthcare professionals should prioritize patient safety, follow proper sterilization and infection control measures, and avoid any actions that could potentially harm the patient.

3. **Veracity:** Veracity refers to the duty of healthcare professionals to be truthful and honest with patients. In the operation theatre, this involves providing accurate and transparent information about the surgical procedure, risks, benefits, and alternatives. Patients should be fully informed to make autonomous decisions about their healthcare.

4. **Autonomy:** Autonomy recognizes the right of patients to make decisions about their own healthcare. In the operation theatre, respecting patient autonomy means obtaining informed consent before the surgical procedure. Patients should be provided with all necessary information, risks, and alternatives, allowing them to make voluntary and informed decisions about their treatment.

5. **Fidelity:** Fidelity refers to the principle of loyalty and trustworthiness in the patient-provider relationship. In the operation theatre, healthcare professionals should uphold fidelity by honoring

their commitments to patients, maintaining patient confidentiality, and acting in the best interest of the patient at all times.

6. Justice: Justice involves the fair and equitable distribution of healthcare resources and services. In the operation theatre, this principle requires healthcare professionals to allocate resources appropriately, ensuring equal access to surgical care for all patients. Justice also entails avoiding any form of discrimination or bias in the provision of surgical services.

pillars of medical ethics which are defined as follows:

1. Patient Autonomy: Respecting the autonomy and informed consent of the patient is critical. It is important to ensure that patients understand the risks, benefits, and alternatives to the surgery, allowing them to make informed decisions about their own healthcare.

2. Confidentiality: Healthcare professionals should maintain patient confidentiality and privacy throughout the surgical process. This includes protecting patient information and only sharing it with those directly involved in the patient's care.

3. Beneficence: Healthcare professionals have a duty to act in the best interest of the patient. This involves providing the highest quality of care, minimizing harm, and promoting the patient's well-being.

4. Non-maleficence: The principle of "do no harm" emphasizes the importance of avoiding actions that could potentially harm the patient during the surgical procedure. This includes ensuring proper sterilization and infection control measures are followed.

5. Justice: Ensuring fairness and equitable access to surgical care is an important ethical consideration. Healthcare professionals should strive to allocate resources appropriately and avoid any form of discrimination or bias.

6. Respect for colleagues: Operating room personnel should treat their colleagues with respect and professionalism. Collaboration and effective communication are essential for providing safe and efficient surgical care

- **Autonomy** – respect for the patient’s right to self-determination
- **Beneficence** – the duty to ‘do good’
- **Non-Maleficence** – the duty to ‘not do bad’
- **Justice** – to treat all people equally and equitably.

Ethical standards in operation theatre

1. decision-making capacity

Patients with decision-making capacity (DMC) have the right to refuse any type of surgery, even one that is life-sustaining. They also have the right to choose between treatment options, based on the principle of informed consent

2. right to refuse treatment

Every competent adult has the right to refuse unwanted medical treatment. This is part of the right of every individual to choose what will be done to their own body, and it applies even when refusing treatment means that the person may die.

3. Withholding and withdrawing treatment

Thanks to recent breakthroughs in medical science and technology, doctors now have available a variety of machines, devices and treatments to keep people alive when an important body system stops working properly. Examples include breathing machines, feeding tubes, CPR and dialysis machines

4. Consent to the patient

Before any surgical procedure is done to the patient, he/she must consent that they are ready for the procedure and also ready for any outcome.

5. Confidentiality to the patient

Any information provided by the patient must be confidential between the surgical operator and the patient beyond which no one should know.

MAINTAINING OPERATION THEATRE

- An ideal operative environment can be defined as an aseptic and undoubtedly a safe place which is devoid of any harmful microbial organisms and pollutants.
- A protective and clean atmosphere in the operation theatre can ensure in maintaining vital functions, prevent dangerous secondary infections and thereby promote healing without any complications.
- The establishment of an operation theatre needs very specific designing, planning, and execution to carry out diagnostic or investigative, therapeutic or surgical procedures.
- The choice of surface finishes, flooring, and layout should be designed for easy infection control.
- The ceiling panels, rear panels, and other fitting surfaces should be compatible, smooth and easy to maintain along with being fire-resistant, water-proof and durable.
- A proper ventilation and air flow system are intended to maintain the optimum air quality thereby assuring the safety of the persons within. It helps to keep the accurate temperature, pressure, and humidity that is ideal and efficient.
- Besides the quality of an operating theatre, the working condition of the instruments, machinery, pre-operative, operative and post-operative equipment should be kept clean and sterile.
- Pre and postoperative sterilization of the various zones in the operation theatre can assure reliability and aid in the uneventful healing of the patient.

Basic anatomy and physiology

Basic nursing skills.

Nursing Skills Checklist

1. Decision Making

Nurses make decisions when the stakes are high. Therefore, their ability to make

informed, cool-headed choices is crucial. They must know how to gather information, prioritize outcomes, and react to urgent and life-threatening conditions effectively. When nurses make decisions, they must take ethical factors into consideration and be able to explain their choices to doctors, patients, families, and other healthcare professionals.

2. Mathematics

Mathematics includes quick and accurate manipulation of numbers. Nurses must know how to work with the metric system and take measurements of a variety of data such as height, weight, and bodily outputs. They must be able to calculate the rates of change for vital signs, manage IV drip rates, and discern appropriate medication doses. This means a good working knowledge of fractions and ratios, arithmetic, formulas, conversion rates, and time calculations.

3. Patience

Nurses see people at their worst. Patients are often scared, in pain, disoriented, or simply unable to function at their full capacity. The ability to decelerate and approach situations calmly means nurses can provide a higher standard of care and avoid burnout. Successful nurses learn to cultivate immersive attention, not take things personally, and approach problems for a long-range angle.

4. Compassion

Compassion is inextricably linked with patience. It manifests as the practical application of kindness. This practice is sometimes difficult to pin down, but it may look like some of the following:

1. Validating patients' concerns and honoring their requests
2. Listening
3. Considering other points of view
4. Making eye contact

5. Facing patients while talking
6. Using a soft vocal tone
7. Smiling
8. Treating patients as equals
9. Nurses who show compassion improve patient satisfaction and keep their work in line with the spirit of their profession.

5. Non-Judgmental

Staying non-judgmental is part of providing an ethical standard of care. Nurses must build trusting relationships with their patients and are obligated to preserve a patient's dignity and independence as much as possible. When nurses are non-judgmental, they are actively sensitive to the physical, psychological, cultural, and spiritual needs of patients. They acknowledge their patients' vulnerabilities and protect them from all unnecessary emotional harm.

6. Physical Endurance

Shifts can last between eight and 12 hours, during which nurses are almost constantly on their feet. Patients often require assistance with walking and movement. Responding to emergencies is a given. Nurses must be able to do all the following for sustained periods of time:

1. Standing
2. Sitting
3. Lifting
4. Carrying
5. Pushing/pulling
6. Bending
7. Reaching
8. Kneeling
9. Twisting

10. Fine motor skillsIn addition, nurses must possess acuity in all their senses, including visual, aural, and tactile.

7. **Observation**

Nurses are conduits. As professionals on the front lines of care, their ability to effectively provide accurate information to doctors, other nurses, patients, and family members is crucial. Successful nursing observation is both evidence-based and vigilant. It informs the work of a host of concepts vital to the field, such as:

1. Listening to patients
2. Practicing nursing procedures effectively
3. Gaining knowledge and understanding of people and situations
4. Communication
5. Patience
6. Teamwork
7. Paperwork and record keeping
8. Empathy

8. **Team-Oriented**

Nurses collaborate constantly. Successful teamwork in nursing means the ability to communicate effectively, stay adaptable, and find ways to get along with others.

Some practical ways that healthcare workers can practice being team-oriented include:

- Staying mindful of how personal values and assumptions affect interpersonal interaction
- Resolving conflicts
- Staying aware of how changes impact team members
- Providing mutual support
- Observing changes in team member behaviors

- Identifying measurable goals and working together to achieve them Like all other skills, successful teamwork leads to increased patient safety and satisfaction, reduces clinical errors, and prevents burnout.

9. **Communication**

Communication refers to verbal, non-verbal, and written means. The skill of effective communication is important since;

1. Allows nurses to build good therapeutic relationships
2. Enables nurses to provide care and comfort
3. Provides a way to administer evidence-based physical interventions
4. Provides avenues to teach
5. Enables manipulation of the nursing environment
6. Helps nurses adopt complementary health practices. Furthermore, when nurses present healthy communication, it can help patients to:

10.) **Dedication**

When nurses possess dedication, they understand they are in their careers for the long haul. This kind of passion for the work means they can continue to make decisions that both improve patient care and help them grow as professionals.

IDENTIFICATION AND PREPARATION OF PATIENT FOR SURGERY

Identification is defined as the act of determining who someone or what something is.

Preparation is the action or process of making something ready for use or service or of getting ready for some occasion, test, or duty.

IDENTIFICATION OF PATIENT.

The patient should have at least two corroborating patient identifiers as evidence to confirm identity.

- ✓ The use of two patient identifiers improves the reliability of the patient identification process and decreases the chance of performing the wrong procedure on the wrong patient.
- ✓ The use of two patient identifiers is necessary in the instances of a name patient alert because two (or more patients) have the same name that can be spelled the same or pronounced the same.

Examples of acceptable patient identifiers include;

- A. Name
- B. Assigned identification number
- C. Telephone number
- D. Date of birth
- E. Social security number
- F. Address
- G. Photograph

- ✓ The patient's room number should not be used as a patient identifier; room numbers are not person-specific identifiers, since patients can be moved from room to room.

All patients undergoing a surgical procedure should wear an identifying marker.

1. Identification markers on the patient will prevent wrong patient surgery.
2. Identification markers on the patient will prevent wrong-procedure and wrong-site surgery.

3. Identification markers can include :

- A. Wristband as identification bracelet
- B. Wristband with unique bar-coded patient identifier
- C. Radio frequency identification (RFID) marker

Recommended times for verification of patient identity and surgical procedures:

- (1) When the surgery is scheduled
- (2) When the patient is admitted to the healthcare facility
- (3) Anytime the patient is transferred to another caregiver
- (4) Prior to sedation
- (5) Prior to the patients entry into the operating room

Recommendations for the identification of the conscious, competent patient prior to the start of the surgical procedure :

- (1) The surgical technologist should address the patient using his/her full name and introduce himself/herself, including job title or position. This will aid in lessening the anxiety of the patient.
- (2) Patient should be asked to say his/her name, the surgical procedure to be performed, and location of the operation.
- (3) The patient's name and hospital-assigned identification number on the surgery schedule and transfer slip should correspond with the information on the patient's wristband.
- (4) The information on the patient's wristband should correspond with the information in the patient's chart.
- (5) Verify that the procedure listed and described on the informed consent in the patient's chart is the same procedure that the patient verbally stated.
- (6) Confirm that the correct procedure is on the operating room schedule.

Recommendations for the identification of the mentally incapacitated patient :

- (1) Verify that the correct patient is being taken to the operating room by asking a family member or designated representative the patient's name.
- (2) Verify the information on the patient's wristband is the same as the information in the patient's chart.
- (3) The patient's name and hospital-assigned identification number on the surgery schedule and transfer slip should correspond with the information on the patient's wrist band.
- (4) Verify that the procedure listed and described on the informed consent in the patient's chart is the same procedure listed on the surgery schedule.
- (5) Confirm with the family member or designated representative the procedure that is expected to be performed, as well as location of the operation and verify this matches up with the informed consent.

The following are recommendations for the identification of a minor patient :

- (1) Complete the recommendations for an alert, oriented patient.

(2) Confirm the minor patient's name with the parent or legal guardian, the procedure to be performed and location of the operation.

Preparing for Surgery

Preparations for surgery depend on the diagnosis of a patient. However, if it involves undergoing general anesthesia, the patient may be asked to do the following:

- ✓ Stop drinking and eating for a certain period of time before the time of surgery
- ✓ Bathe or clean, and possibly shave the area to be operated on
- ✓ Undergo various blood tests, X-rays, electrocardiograms, or other procedures necessary for surgery
- ✓ Sometimes a patient may be asked to take an enema the evening before surgery, to empty the bowels.
- ✓ Do not wear makeup the day of surgery
- ✓ Do not wear nail polish
- ✓ Do not wear your eye contacts
- ✓ Leave valuables and jewelry at home
- ✓ Advise the medical staff of dentures or other prosthetic devices you may be wearing

To make their experience more comfortable and efficient, patients are advised to bring the following:

- ✓ Loose-fitting clothes to wear
 - ✓ Social Security card number
 - ✓ Insurance information
 - ✓ Medicare or Medicaid card.
-
- Reduce anxiety
 - Advice on the importance of the surgery
 - Assure the patient that the surgery will be successful.

Preparation of the patient in the pre-surgical holding room

1. Place a warm blanket on the patient, verify the patients identification by name and date of birth and notify the individual at the surgery control desk that the patient is in the holding area.
2. Verify the surgical procedure, site and surgeon verbally with the patient or family as appropriate. The surgical site is validated by observing the indelible ink of the surgeon's initials. Hospital policy is followed for identification of surgical sites that cannot be marked.
3. Review the patients chart and EMR for completeness.
 - Medical history and physical examination
 - Laboratory reports
 - Consent forms and documentation of consents which include; informed consent, general consent and anaesthesia consent.
4. Measure the vital signs and blood pressure.
5. Verify allergies and medication history
6. Check skin tone and integrity
7. Note the patients mental state
8. Put a cap on the patients head to protect his or her hair for purposes of asepsis and help minimize hypothermia. patients who are held are require to wear a head covering to prevent heat loss and scalp dander shed.
9. Notify the individual at the surgery control desk when the patient is ready for transport to the operation room. The patient must be under constant observation by the patient care staff until transported from the surgical department to another patient care unit or discharged.

MAINTAINING OPERATION THEATRE RECORDS

Maintaining detailed and accurate records in an operation theatre (OT) is crucial for ensuring patient safety, facilitating continuity of care, and complying with regulatory requirements. Here's a comprehensive guide on maintaining operation theatre records:

Patient Information:

Collect comprehensive demographic information, including patient's full name, age, gender, address, and contact details.

Document relevant medical history, including past surgeries, allergies, medications, and pre-existing conditions.

Ensure accuracy in recording patient identifiers to avoid errors in patient identification during surgical procedures.

Pre-operative Assessment:

Perform a thorough pre-operative assessment, including physical examination findings, laboratory test results, and anesthesia evaluation.

Document pre-operative consultations, discussions, and consent forms signed by the patient or legal guardian.

Record any special instructions or considerations related to the patient's condition that may impact the surgical procedure.

Surgical Plan:

Document the planned surgical procedure, including the intended approach, technique, and anticipated duration.

Record any modifications or deviations from the original surgical plan discussed or decided upon intraoperatively.

Include details of any pre-operative preparations, such as patient positioning, anesthesia induction, and surgical site marking.

Intraoperative Records:

Maintain a detailed log of intraoperative events, including start and end times of surgery, personnel involved, and surgical findings.

Document anesthesia management, including medications administered, vital signs, and responses to anesthesia.

Record intraoperative complications, unexpected events, or deviations from the expected course of surgery.

Surgical Instruments and Supplies:

Keep an inventory of surgical instruments, implants, and supplies used during each procedure.

Document the usage of disposable items and ensure proper disposal according to institutional protocols.

Track the sterilization status of reusable instruments and equipment, including sterilization dates and validation records.

Anesthesia Records:

Maintain detailed anesthesia records, including pre-operative assessment, induction, maintenance, and emergence phases.

Document anesthesia drugs administered, dosages, routes of administration, and any adverse reactions observed.

Record vital signs monitoring data, including blood pressure, heart rate, respiratory rate, oxygen saturation, and temperature.

Post-operative Care:

Document post-operative instructions, medications prescribed, and anticipated recovery milestones.

Record post-anesthesia recovery assessments, including level of consciousness, pain scores, and vital signs stability.

Ensure clear communication of post-operative care plans to the patient, caregivers, and post-operative care team.

Pathology Reports:

Include pathology reports for specimens collected during surgery, documenting histopathological findings and interpretations.

Ensure timely communication of pathology results to the surgical team for further management decisions.

Consent Forms and Documentation:

Maintain signed consent forms for surgical procedures, anesthesia, blood transfusions, and other interventions requiring patient consent.

Document any changes or updates to consent forms and ensure patient understanding and agreement.

Incident Reporting and Quality Improvement:

Implement a system for reporting and documenting adverse events, near misses, and incidents occurring in the operation theatre.

Conduct regular audits and reviews of operation theatre records to identify areas for improvement and opportunities for enhancing patient safety and quality of care.

Legal and Regulatory Compliance:

Ensure compliance with institutional policies, regulatory requirements, and accreditation standards governing operation theatre recordkeeping.

Safeguard patient confidentiality and privacy in accordance with applicable laws and regulations, such as Health Insurance Portability and Accountability Act (HIPAA) guidelines.

Documentation Management:

Maintain organized and legible operation theatre records, either in electronic health record (EHR) systems or paper-based formats.

Standardize documentation practices and ensure consistency in recording essential information across all operation theatre records

1. Documentation: All relevant information related to the surgical procedure should be documented accurately and legibly. This includes pre-operative assessments, consent forms, surgical notes, anesthesia records, and post-operative care details.

2. Standardized Forms: Using standardized forms or electronic systems for recording operation theatre data can help ensure consistency and completeness of documentation. These forms should capture essential information such as patient demographics, surgical details, medications, and any complications or adverse events.

3. Timeliness: Operation theatre records should be completed in a timely manner. It is important to document events and interventions promptly to ensure accuracy and avoid potential memory lapses.

4. Signature and Authentication: All entries in the operation theatre records should be signed or electronically authenticated by the responsible healthcare professional. This verifies the accuracy and accountability of the information documented.

5. Privacy and Confidentiality: Operation theatre records contain sensitive patient information and should be treated with strict confidentiality. Access to these records should be limited to authorized personnel only, following appropriate security measures to protect patient privacy.

6. Retention and Storage: Operation theatre records should be retained according to local regulations and institutional policies. These records should be stored securely, ensuring protection against loss, damage, or unauthorized access. Digital storage systems should have appropriate backup and security measures in place.

7. Quality Assurance: Regular audits and reviews of operation theatre records can help identify any documentation deficiencies or areas for improvement. This can contribute to enhancing the quality and accuracy of the records and ultimately improve patient care.

8. Legal and Ethical Considerations: Operation theatre records may serve as legal documents in case of any medico-legal issues. Therefore, it is crucial to maintain complete, accurate, and unbiased records, adhering to ethical and legal guidelines.

DISPOSAL OF THEATRE WASTE

Types of waste in operation theatre

- **Infectious waste:** waste contaminated with blood and other bodily fluids (e.g. from discarded diagnostic samples), cultures and stocks of infectious agents from laboratory work (e.g. waste from autopsies and infected animals from laboratories), or waste from patients with infections (e.g. swabs, bandages and disposable medical devices);
- **Pathological waste:** human tissues, organs or fluids and body parts.
- **Sharps waste:** syringes, needles, disposable scalpels and blades, etc.;
- **Chemical waste:** for example, disinfectants, sterility and heavy metals contained in medical devices (e.g. mercury in broken thermometers)
- **Pharmaceutical waste:** expired, unused and contaminated drugs and vaccines;
- **Cytotoxic waste:** waste containing substances with geotaxis properties (i.e. highly hazardous substances that are, mutagenic, teratogenic or carcinogenic), such as cytotoxic drugs used in cancer treatment and their metabolites;
- **Radioactive waste:** such as products contaminated by radionuclides including radioactive diagnostic material or radiotherapeutic materials; and
- **Non-hazardous or general waste:** waste that does not pose any particular biological, chemical, radioactive or physical hazard. (tissue, food particles, papers)

Waste disposal methods

Incineration – This is the high-temperature burning (rapid oxidation) of a waste. Used for burning infectious waste , pathological waste and sharps.

Cat	Type of Container	Type of waste	Treatment/ Disposal options
	Non-chlorinated plastic bags	(a) Human Anatomical Waste (b) Animal Anatomical Waste (c) Soiled Waste (d) Expired or Discarded Medicines (e) Chemical Waste (f) Micro, Biotechnological and other clinical lab waste (g) Chemical Liquid Waste	Incineration or Plasma pyrolysis or deep burial
	Non-chlorinated plastic bags or containers	Contaminated Waste (Recyclable) tubing, bottles, intravenous tubes and sets, catheters, urine bags, syringes (without needles) and gloves	Autoclaving/ microwaving /hydroclaving and then sent for recycling
	(Translucent) Puncture, Leak, tamper proof containers	Waste sharps including Metals	Auto or Dry Heat Sterilization followed by shredding or mutilation or encapsulation
	Cardboard boxes with blue colored marking	Glassware	Disinfection or autoclaving, microwaving, hydroclaving and then sent for recycling

HANDLING OF OPERATION THEATRE WASTE

When we think of people who handle medical waste, workers at healthcare facilities generally come to mind. However, there is a whole array of other folks, from transporters to landfill staff, who could also potentially be exposed to medical waste. As we are all aware, no matter what precautions we take, there is always the possibility of spills or other exposure.

- If you encounter a red bag, red sharps container, or something marked with the biohazard symbol; or sharps, blood or blood-containing material, human or animal parts; or other material that appears to be medical waste, **treat it as if it is hazardous**. Inform your supervisor before covering it or handling it in any way. Your supervisor may want to reject the waste, and, if so, will contact the hauler, or the person who disposed the waste, to come pick it up.
- Wear PPE under circumstances in which you might be exposed. Boots should have steel toes and puncture-resistant soles, and 6 inch lace-up boots provide added protection for the ankles. Gloves should be worn whenever the hands may come in contact with hazards; leather provides better protection than rubber against punctures. Arms and other skin surfaces should be covered whenever the skin might be exposed to infectious agents. Safety glasses and hard hats may also provide protection to the head and face from splashes.
- Special equipment, such as respirators, face shields, dust masks, boot covers, or impervious clothing may be necessary if a spill occurs, if splashing or splattering is expected, or if another unusual hazard arises.
- Cover all cuts, abrasions, and other areas of non intact skin while on duty.
- Avoid physical contact with medical wastes, whether they are in red bags or sharps containers or not. If you are required to move these items—for example, to separate them for pickup—use a shovel or other implementation.
- Never handle any wastes with your bare hands.
- Be aware of the possible presence of medical wastes when handling all wastes and when cleaning all machinery and equipment, and try to avoid contact. In particular, be aware that sharps can become stuck in the wheels and tracks of vehicles, and can pose hazards to operators and maintenance workers.
- Where possible, use some implement or cleaning method other than your hands. If you do have to use your hands to clean or maintain equipment, make sure that you are wearing gloves that minimize the chance of being cut.
- Never reach where you cannot see.

- Large pieces of broken glass should be removed with brooms and disposed of in a separate container. Small pieces can be removed with tongs. Glass should never be removed with fingers.
- Avoid handling personal items, like pens and combs, while wearing gloves.
- Always wash your hands after removing your gloves, even if the gloves have not been cut or punctured.
- Always wash before eating, drinking, smoking, or putting anything in your mouth, and before leaving work.
- Remove clothing, boots, and gloves that have been in contact with infectious waste as soon as possible, taking care to avoid contact with exposed skin surfaces.
- Use gloves to remove other items of clothing.
- Place contaminated PPE in closable, leakproof bags or containers for disposal or decontamination.
- If you come in contact with infectious waste (for example, if you are splashed with blood or blood-containing bodily fluids), wash your hands and any exposed skin thoroughly in warm water and soap, or in waterless antiseptic cleaner, if soap and water are unavailable.
- Report any on-the-job exposure to blood or other bodily fluids promptly and get medical attention.